

NETWORKED CONTROL

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PART III: NETWORKED

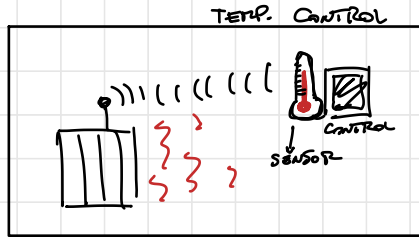
CONTROL SYSTEMS

The present notes are not to be considered as a substitute of the personal notes or of the official lecture notes freely available on webeep.

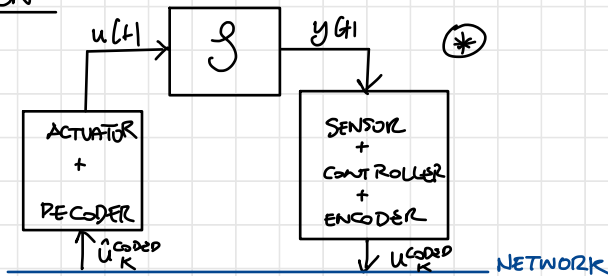
1. CONTROL UNDER RATE LIMITATIONS

A) INTRODUCTION

• EXAMPLE



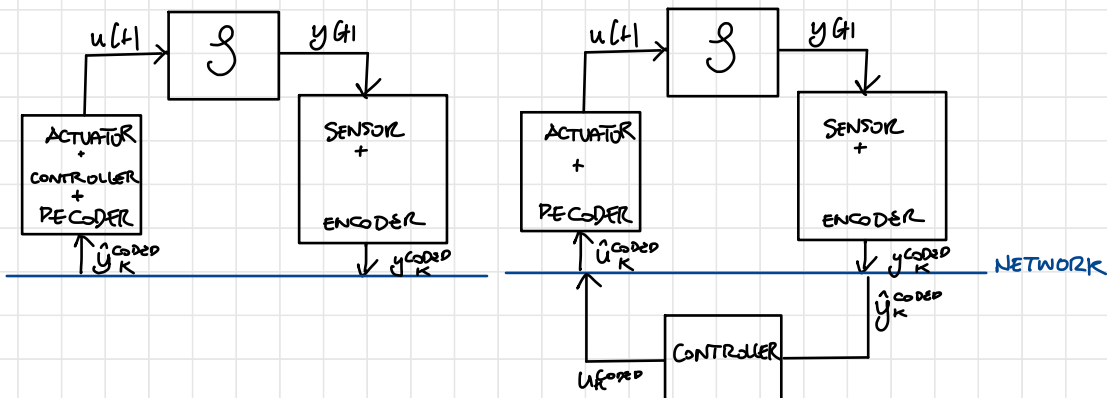
• CONFIGURATION



IF, FOR EXAMPLE, $N_B = 1$ BITS/PACKET, THE INPUT CAN ASSUME $N = 2^{N_B} = 2$ VALUES, FOR EXAMPLE

$$u_k = \begin{cases} u_{\min} & \text{if } \hat{u}_k^{\text{coded}} = 0 \\ u_{\max} & \text{if } \hat{u}_k^{\text{coded}} = 1 \end{cases}$$

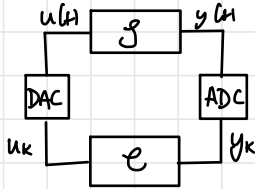
OTHER POSSIBLE CONFIGURATIONS :



FOR SIMPLICITY, IN THIS LECTURE WE WILL MAKE REFERENCE ONLY TO CONFIGURATION (*)

• MAIN DIFFERENCE WITH RESPECT TO DIGITAL CONTROL

- DIGITAL CONTROL SCHEME: $u_k \in \mathbb{R}$



- IN CASE OF RATE LIMITATIONS u_k CAN TAKE JUST A FINITE NUMBER OF VALUES! $u_k \in \mathcal{U}$

$\mathcal{U}$: SET OF ADMISSIBLE INPUTS

$N = 2^{N_B}$: NUMBER OF ADMISSIBLE INPUTS



YOU CANNOT APPLY A CONTROL LAW OF TYPE $u_k = K y_k$

• ASSUMPTIONS ON THE COMMUNICATION CHANNEL

- The channel has a capacity $R_{\max}$

- We use a packet network



Each packet has N_B bits

REMARK: normally each packet has N_B bits
but only N_B^i includes essential information

IN THIS CASE

$$R_{\max}^{\text{EFF}} = R_{\max} \cdot \frac{N_B^i}{N_B} \Rightarrow \text{N. of information bits/second}$$

↳ effective capacity

For simplicity we will consider $N_B^2 = N_B$

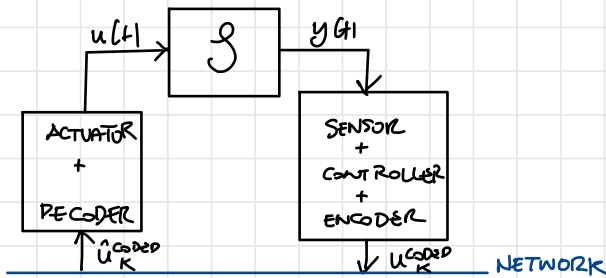
$$\Rightarrow \underline{R_{max}^{eff} = R_{max}}$$

- THE RATE AT WHICH THE BITS ARE TRANSMITTED IS $R < R_{max}$
BY DESIGN

- THE COMMUNICATION LINK IS "IDEAL"

- NO DELAY
- NO NOISE
- NO DROPOUT

B) PROBLEM - FIXED R ? HOW TO CHOOSE R , SAMPLING TIME h , N_B ?



OPTIMAL SITUATION : - FOR MAKING THE CONTROL SYSTEM REACTIVE
 h SHOULD BE AS SMALL AS POSSIBLE

- FOR MAKING THE CONTROLLER "FINE" AND
EFFECTIVE N_B SHOULD BE AS LARGE
AS POSSIBLE

HOWEVER $T_{frame} = \frac{N_B}{R} \leq h$ (IF POSSIBLE $\ll$)

$$\Downarrow$$
$$\boxed{\frac{N_B}{h} \leq R}$$

- IF YOU FIX h SMALL $\Rightarrow N_b \leq R h \Rightarrow$ LIMITATION ON THE UPPER BOUND OF N_b !
- IF YOU FIX N_b LARGE $\Rightarrow h \geq N_b/R \Rightarrow$ LIMITATION ON THE LOWER BOUND OF h !

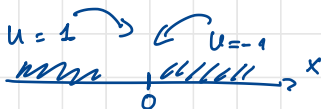
$\Downarrow$
 Performance limitations!

$$? \leq \frac{N_b}{h} \leq R$$

c) EXAMPLES $\dot{x} = ax + bu$, $N_b = 1 \text{ bit} \Rightarrow u_k = \begin{cases} 1 \\ -1 \end{cases}$
 $(a = -1, 0.5, 1)$ $h = 1 \text{ s}$

BINARY CONTROL LAW: $u_k = \begin{cases} 1 & \text{IF } x_k < 0 \\ -1 & \text{IF } x_k \geq 0 \end{cases}$

IN DISCRETE-TIME



$$x_{k+1} = f x_k + g u_k, \quad f = e^{ah}, \quad g = -\frac{b}{a}(1 - e^{ah})$$

UNDER THE CONTROL LAW:

(IF $a \neq 0$)

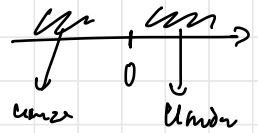
$$x_{k+1} = \begin{cases} f x_k + g & \text{IF } x_k < 0 \\ f x_k - g & \text{IF } x_k \geq 0 \end{cases}$$

$$1) \quad a = -1 \quad \Rightarrow \quad f = 0.37 \quad g = 0.63$$

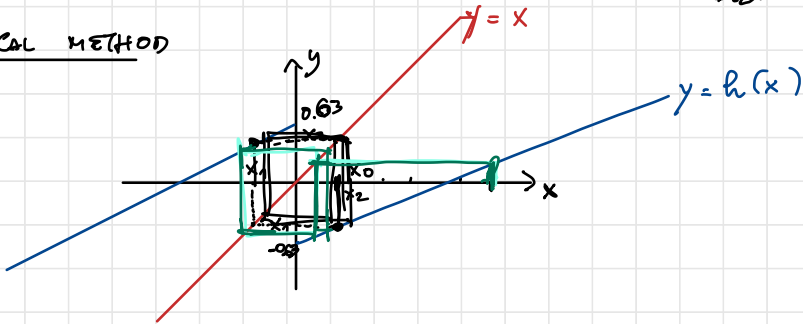
$$x_{k+1} = 0.37 x_k + 0.63 u_k$$

$$u_k = \begin{cases} 1 & \text{if } x_k < 0 \\ -1 & \text{if } x_k \geq 0 \end{cases}$$

$$h(x_k) = \begin{cases} 0.37 x_k + 0.63 & \text{if } x_k < 0 \\ 0.37 x_k - 0.63 & \text{if } x_k \geq 0 \end{cases}$$



GRAPHICAL METHOD



$\forall x_0$

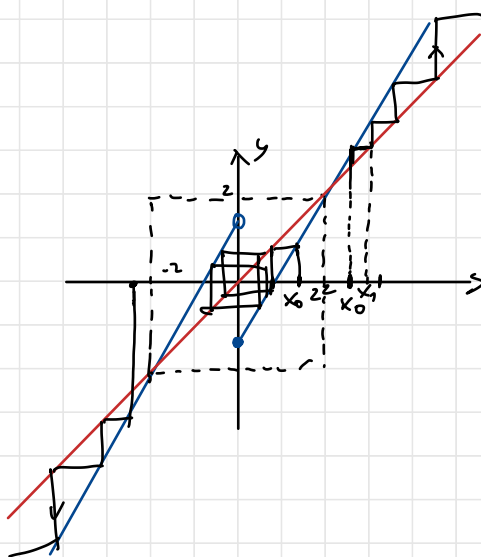
$$x_k \rightarrow [-0.63, 0.63]$$

$$2) \quad a = 0.5 \quad \underline{\text{unstable!}}$$

$$f = 1.65 \quad g = 1.3$$

$$x_{k+1} = 1.65 x_k + 1.3 u_k$$

$$h(x_k) = \begin{cases} 1.65 x_k - 1.3 & x_k \geq 0 \\ 1.65 x_k + 1.3 & x_k < 0 \end{cases}$$



$$o \quad \underline{|x_0| > 2} \quad |x_k| \rightarrow +\infty$$

$$o \quad \underline{|x_0| = 2} \quad |x_k| = 2 \quad \forall k$$

$$o \quad \underline{|x_k| < 2}$$

$$|x_k| < 2 \quad x_k \in [-2, 2]$$

$$x_k \rightarrow [-1.3, 1.3]$$

Both $[-2, 2]$ and $[-1.3, 1.3]$ are positively invariant sets

$$3) \quad \underline{a = 1}$$

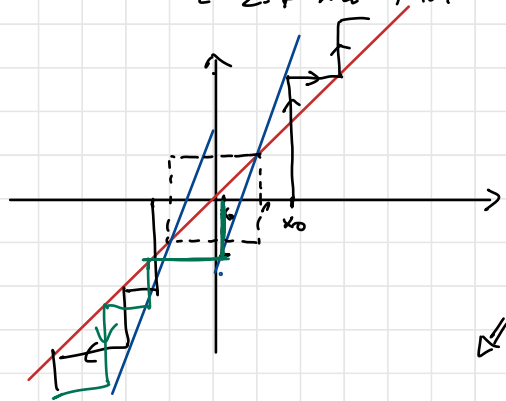
$$f = 2.7$$

$$g = 1.7$$

$$u = \{-1, 1\}$$

$$x_{k+1} = 2.7 x_k + 1.7 u_k$$

$$\begin{cases} 2.7 x_k - 1.7 & x_k \geq 0 \\ 2.7 x_k + 1.7 & x_k < 0 \end{cases}$$



$$|x_k| > 1 \quad |x_k| \rightarrow +\infty$$

$$|x_k| < 1 \quad |x_k| \rightarrow +\infty$$

$$|x_k| = 1 \quad |x_k| = 1$$



INSTABILITY

REMARKS

DIFFERENCES WITH DIGITAL CONTROL (SINGLE-INPUT CASE)

DIGITAL CONTROL

$$\textcircled{1} u_k \in \mathbb{R}$$

• CONTROLLABILITY: $\forall x_0 \in \mathbb{R}^n$
 $\exists u_0, u_1, \dots, u_R$ s.t.
 $x_{R+1} = 0$

• STABILIZABILITY: $\forall x_0 \in \mathbb{R}^n$
 $\exists u_0, u_1, u_2, \dots$ s.t.
 $x_k \rightarrow 0, k \rightarrow \infty$

• We can use $u_k = K x_k$

CONTROL UNDER RATE LIMITATIONS

$$\textcircled{1} u_k \in \mathcal{U} \text{ DISCRETE FEASIBLE INPUT SET}$$

$$\mathcal{U} = \{\bar{u}_0, \dots, \bar{u}_{N-1}\} \quad N = 2^{N_B}$$

$$\forall x_0 \in \text{SET} \quad \exists u_0, u_1, u_2, \dots \text{ s.t.}$$

$$x_k \in \text{SET} \quad \text{*ROUND ORIGIN *K}$$



What type of "stabilizability" notion do we need?

? GIVEN R , CAN WE SELECT N_B, h SUCH THAT $N_B/h < R$ AND SUCH THAT THE SYSTEM CAN BE CONTROLLED?

Control law cannot be proportional to x_k
 What kind of control law could we apply?

D) BOUNDABILITY

DEFINITION: AN LTI SYSTEM

$$\dot{x} = Ax + Bu$$

with an input set U (discrete, with $N = 2^{N_B}$) is BOUNDABLE

if $\exists$ bounded set S and an open set $M \subseteq S$ such

that, $\forall x_0 \in M$, $\exists$ control sequence $u_0, u_1, u_2, \dots, u_K \in U$

such that $x(t) \in S \quad \forall t$, if we apply the

control sequence in a sample-and-hold way with sampling time h .

? How can we guarantee BOUNDABILITY?

? What kind of control law can we use?

E) SCALAR CASE

• SYSTEM $\dot{x} = ax + bu$ ($b > 0$ for simplicity)

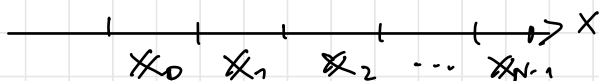
• INPUT VALUES $u_k \in \{\bar{u}_0, \bar{u}_1, \dots, \bar{u}_{N-1}\}$, $N = 2^{N_B}$

$$u \in [u_{\min}, u_{\max}] \rightarrow \text{Bounded!}$$

• INPUT IS APPLIED IN A SAMPLE-AND-HOLD WAY WITH SAMPLING TIME h

$$\hookrightarrow x_{k+1} = f x_k + g u_k \quad \rightarrow \text{Discrete time evolution}$$

• CONTROL LAW



The control law must be of the type :

$$u_k = \bar{u}_i \quad \text{IF} \quad x_k \in X_i$$

$\downarrow$
 $= S(x_k)$ SELECTION FUNCTION

THEOREM

There exists a finite set $\mathcal{U}$ s.t. the system is BOUNDABLE

if and only if $\frac{N_B}{h} \geq a \log_2 *$

$\Rightarrow$ Under condition $*$ $\exists$ a suitable selection function!

Note that, based also on the previous conditions, the tradeoff for $\frac{N_B}{h}$ is

$$a \log_2 e \leq \frac{N_B}{h} < R$$

This implies, for example, that if $R < a \log_2 e$ we can never find N_B and h that make the system boundable!

IN THE EXAMPLES:

1) $a = -1 \Rightarrow a \log_2 e = -1.44 < \frac{N_B}{h} = 1$ OK!

2) $a = 0.5 \Rightarrow a \log_2 e = 0.72 < \frac{N_B}{h} = 1$ OK!

3) $a = 1 \Rightarrow a \log_2 e = 1.44 \not< \frac{N_B}{h} = 1$ NO!

Considering condition $\frac{N_B}{h} \geq a \log_2 e$

Two possibilities $\Rightarrow$

- $\frac{N_B}{h} = a \log_2 e \Rightarrow$ IN THIS CASE WE OPERATE AT THE RATE LIMIT
- $\frac{N_B}{h} > a \log_2 e$

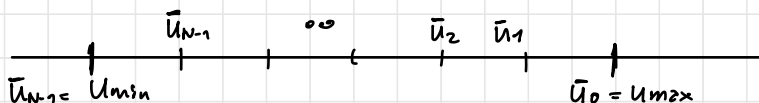
F) SCALAR CASE, WHEN WE OPERATE AT THE RATE LIMIT

When $\frac{Nb}{R} = a \log_2 e$

Remark: WE OPERATE
AT THE RATE LIMIT
ONLY IN CASE $a > 0$

- THE SELECTION FUNCTION IS UNIQUE!
- $U = \{\bar{u}_0, \bar{u}_1, \dots, \bar{u}_{N-1}\}$ WHERE
 $\bar{u}_0, \bar{u}_1, \dots, \bar{u}_{N-1}$ MUST BE EQUALLY
SPACED

We denote $\bar{u}_0 = U_{\max}$, $\bar{u}_{N-1} = U_{\min}$,
and



$$\Rightarrow \bar{u}_i = U_{\max} + (U_{\min} - U_{\max}) \frac{i}{N-1} \quad i=0, \dots, N-1$$

The selection function is:

$$\underline{u_k} = S(x_k) = \begin{cases} \bar{u}_0 = U_{\max} & \text{if } x_k \leq \bar{x}_1 \\ \bar{u}_i & \text{if } \bar{x}_i < x_k \leq \bar{x}_{i+1} \\ \bar{u}_{N-1} = U_{\min} & \text{if } x_k > \bar{x}_{N-1} \end{cases}$$

where:

$$\bar{x}_i = -\frac{b}{a} U_{\max} + \frac{b}{a} (U_{\max} - U_{\min}) \frac{i}{N}$$

The positively invariant set is $[\bar{x}_0, \bar{x}_N]$

Remark: • $S(x_k)$ IS THE ONLY POSSIBLE SELECTION FUNCTION WHEN
 $\frac{Nb}{R} = a \log_2 e$

- IN CASE $\frac{Nb}{R} > a \log_2 e$ THE DEFINED SELECTION FUNCTION
STILL WORKS, BUT IT IS NOT UNIQUE! WE HAVE MORE
POSSIBILITIES (FOR EXAMPLE LOGARITHMIC SPACINGS)

F) HIGH-ORDER SYSTEMS

$$\dot{x} = Ax + Bu$$

• u scalar

• (A, B) controllable

The bound established before still holds, but with the necessary differences.

⇒ The system is boundable if and only if

$$\cancel{a} \log_2 e \leq \frac{NB}{h}$$

$$\operatorname{Re}(\lambda_1) + \operatorname{Re}(\lambda_2) + \dots + \operatorname{Re}(\lambda_k), \text{ where}$$

$\lambda_1, \dots, \lambda_k$ are the eigenvalues of A with
POSITIVE REAL PART

EXERCISES

1)

Consider the following data, regarding some widespread control networks.

	ETHERNET	DEVICENET	CONTROLNET	WIFI
Link length l	2.5 km	100 m	1000 m	150 m
Propagation speed v	$2 \cdot 10^8$ m/s	$2 \cdot 10^8$ m/s	$2 \cdot 10^8$ m/s	$3 \cdot 10^8$ m/s
Bitrate R	1 Gb/s	500 kb/s	5 Mb/s	54 Mb/s
Packet size	<u>1526</u> bytes	8 bytes (message) + 47 bits (overhead)	510 bytes	2346 bytes

N_B	12208	111	4080	18768
T_{FRAME}	$12.2 \mu s$	$222 \mu s$	$816 \mu s$	$348 \mu s$
T_{PROP}	$12.5 \mu s$	$0.5 \mu s$	$5 \mu s$	$0.5 \mu s$
T_{TX}	$24.7 \mu s$	$222.5 \mu s$	$821 \mu s$	$348.5 \mu s$
$t_{min} = T_{TX}$	"	"	"	"
a_{max}	$3.42 \cdot 10^8$	$346 \cdot 10^5$	$3.64 \cdot 10^6$	$37.38 \cdot 10^6$

$$T_{TX} = T_{PROP} + T_{FRAME}$$

$$T_{PROP} = \frac{l}{v}$$

$$T_{FRAME} = \frac{N_B}{R}$$

$$h \geq T_{TX}$$

$$\frac{N_B}{h} \geq a \log_2 e \rightarrow a \leq \frac{N_B}{\underline{h_{min}} \log_2 e} = a_{max}$$

$$2) \quad \hat{x} = x + u$$

$$A) \quad \frac{NB}{h} \geq a \log_2 e = 1.44$$

↓
1

$$B) \quad NB \geq \underbrace{1.44 \times h}_{> 1} \quad \boxed{NB_{min} = 2}$$

REMARK

$$\frac{NB_{min}}{h}$$

$$\begin{matrix} & \nearrow & \\ & > & \\ & & a \log_2 e \\ & & \\ & \searrow & \\ & = & \end{matrix}$$

NOT OPERATING AT THE RATE LIMIT

OPERATING AT THE RATE LIMIT

IN THIS CASE WE OPERATE ABOVE THE RATE LIMIT

$$C) \quad NB = 1 \quad u_k = -\text{sign}(x_k) = \begin{cases} -1 & x_k \geq 0 \\ 1 & x_k < 0 \end{cases}$$

$$h = \log_e 2.5$$

$$x_{k+1} = f x_k + g u_k$$

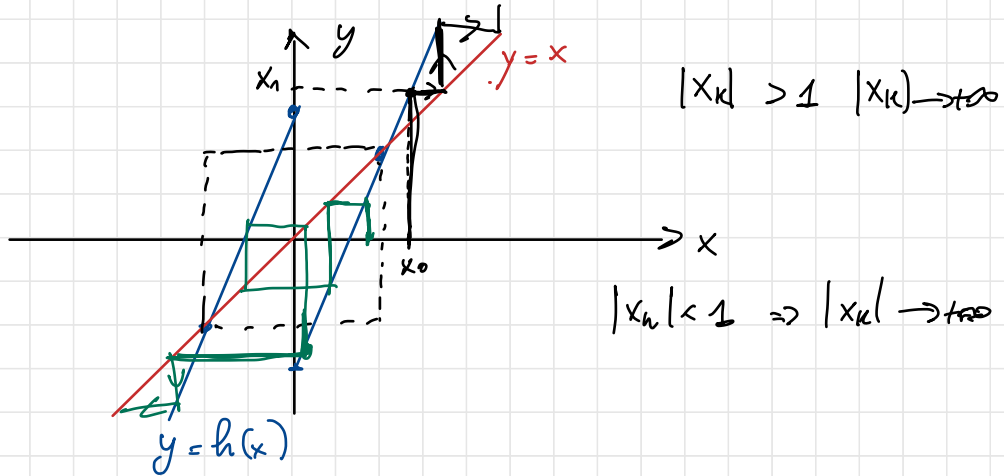
$$f = e^{ah} = e^{\log_e 2.5} = 2.5$$

$$g = -\frac{b}{a}(1 - e^{ah})$$

$$= -(1 - 2.5) = 1.5$$

$$x_{k+1} = 2.5 x_k + 1.5 u_k$$

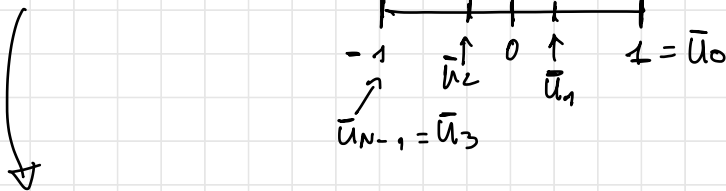
$$| \quad = h(x_k) = \begin{cases} 2.5 x_k - 1.5 & x_k \geq 0 \\ 2.5 x_k + 1.5 & x_k < 0 \end{cases}$$



D) $N_B = 2$ $\ddot{u}$ $N = 2^{N_B} = 4$

$$x_{k+1} = 2.5 x_k + 1.5 u_k$$

$$\bar{u}_i = u_{\max} + (u_{\min} - u_{\max}) \cdot \frac{i}{N-1}$$



$$\bar{u}_0 = u_{\max} = 1$$

$$\bar{u}_1 = u_{\max} + \frac{u_{\min} - u_{\max}}{3} = \frac{2}{3} u_{\max} + \frac{1}{3} u_{\min} = \frac{1}{3}$$

$$\bar{u}_2 = u_{\max} + \frac{u_{\min} - u_{\max}}{3} \cdot 2 = u_{\max} \cdot \frac{1}{3} + \frac{2}{3} u_{\min} = -\frac{1}{3}$$

$$\bar{u}_3 = u_{\min} = -1$$

$$\bar{x}_1 = -\frac{b}{a} u_{\max} + \frac{b}{a} (u_{\max} - u_{\min}) \cdot \frac{1}{N}$$

$$\downarrow$$

$$\bar{x}_0 = -\frac{b}{a} u_{\max} = -1$$

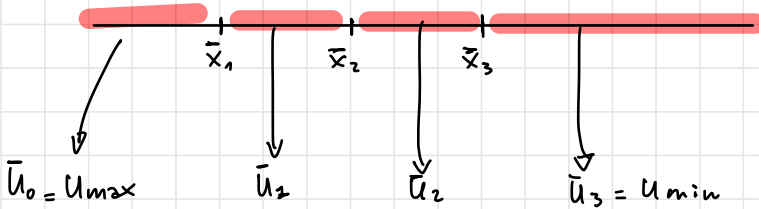
$$\bar{x}_1 = -\frac{b}{a} u_{\max} + \frac{b}{a} (u_{\max} - u_{\min}) \cdot \frac{1}{4}$$

$$= -1 + \frac{2}{4} = -1 + \frac{1}{2} = -\frac{1}{2}$$

$$\bar{x}_2 = -\frac{b}{a} u_{\max} + \frac{b}{a} (u_{\max} - u_{\min}) \cdot \frac{2}{4} = -1 + 1 = 0$$

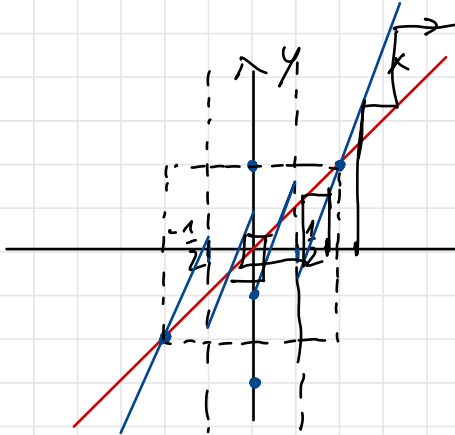
$$\bar{x}_3 = -\frac{b}{a} u_{\max} + \frac{b}{a} (u_{\max} - u_{\min}) \cdot \frac{3}{4} = +\frac{1}{2}$$

$$\bar{x}_4 = -\frac{b}{a} u_{\max} + \frac{b}{a} (u_{\max} - u_{\min}) \cdot \frac{4}{4} = 1$$



$$u_k = S(x_k) = \begin{cases} 1 & x_k < -\frac{1}{2} \\ \frac{1}{3} & -\frac{1}{2} \leq x_k < 0 \\ -\frac{1}{3} & 0 \leq x_k < \frac{1}{2} \\ 1 & x_k \geq \frac{1}{2} \end{cases}$$

$$x_{k+1} = h(x_k) \begin{cases} 2.5x_k + 1.5 & x_k < -\frac{1}{2} \\ 2.5x_k + 0.5 & -\frac{1}{2} \leq x_k < 0 \\ 2.5x_k - 0.5 & 0 \leq x_k < \frac{1}{2} \\ 2.5x_k - 1.5 & x_k \geq \frac{1}{2} \end{cases}$$



$$|x_0| > 1 \Rightarrow |x_k| \rightarrow +\infty$$

$$|x_0| < 1 \Rightarrow |x_k| \text{ Bounded}$$

THE POSITIVELY INVARIANT SET IS $[\bar{x}_0, \bar{x}_N]$, I.E.,
 $[-1, 1]$

$$3) \quad \frac{N_5}{h} \geq \log_2 e = -1.44$$

$$\forall h, N_5$$

$$u_k = s(x_k) = \begin{cases} \bar{u}_0 = u_{\max} & \text{if } x_k \leq \bar{x}_1 \\ \bar{u}_i & \text{if } \bar{x}_i < x_k \leq \bar{x}_{i+1} \\ \bar{u}_{N-1} = u_{\min} & \text{if } x_k > \bar{x}_{N-1} \end{cases}$$

where:

$$\bar{x}_i = -\frac{b}{|a|} u_{\max} + \frac{b}{|a|} (u_{\max} - u_{\min}) \frac{i}{N}$$

The positively invariant set is $[\bar{x}_0, \bar{x}_N]$

$$\bar{u}_0 = 1$$

$$\bar{x}_1 = \frac{1}{2}$$

$$\bar{u}_1 = \frac{1}{3}$$

$$\bar{x}_2 = 0$$

$$\bar{u}_2 = -\frac{1}{3}$$

$$\bar{x}_3 = \frac{1}{2}$$

$$\bar{u}_3 = -1$$

$$u_k = s(x_k) = \begin{cases} 1 & x_k < -\frac{1}{2} \\ \frac{1}{3} & -\frac{1}{2} \leq x_k < 0 \\ -\frac{1}{3} & 0 \leq x_k < \frac{1}{2} \\ 1 & x_k \geq \frac{1}{2} \end{cases}$$

$$\dot{x} = -x + u$$

$$h = \log 2.5$$

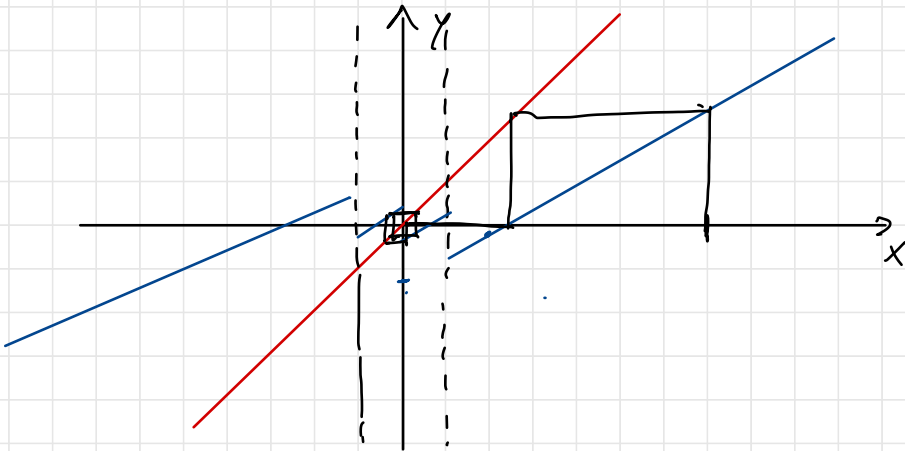
$$x_{k+1} = f x_k + g u_k$$

$$= 0.4 x_k + 0.6 u_k$$

$$f = e^{ah} = e^{-\log 2.5} = \frac{1}{2.5} = 0.4$$

$$g = -\frac{b}{a}(1 - e^{ah}) = 0.6$$

$$= \begin{cases} 0.4 x_k - 0.6, & x_k \geq \frac{1}{2} \\ 0.4 x_k - 0.2, & 0 < x_k < \frac{1}{2} \\ 0.4 x_k + 0.2, & -\frac{1}{2} \leq x_k < 0 \\ 0.4 x_k + 0.6, & x_k < -\frac{1}{2} \end{cases}$$



OTHER CONTROL LAWS CAN BE USED, E.G.,

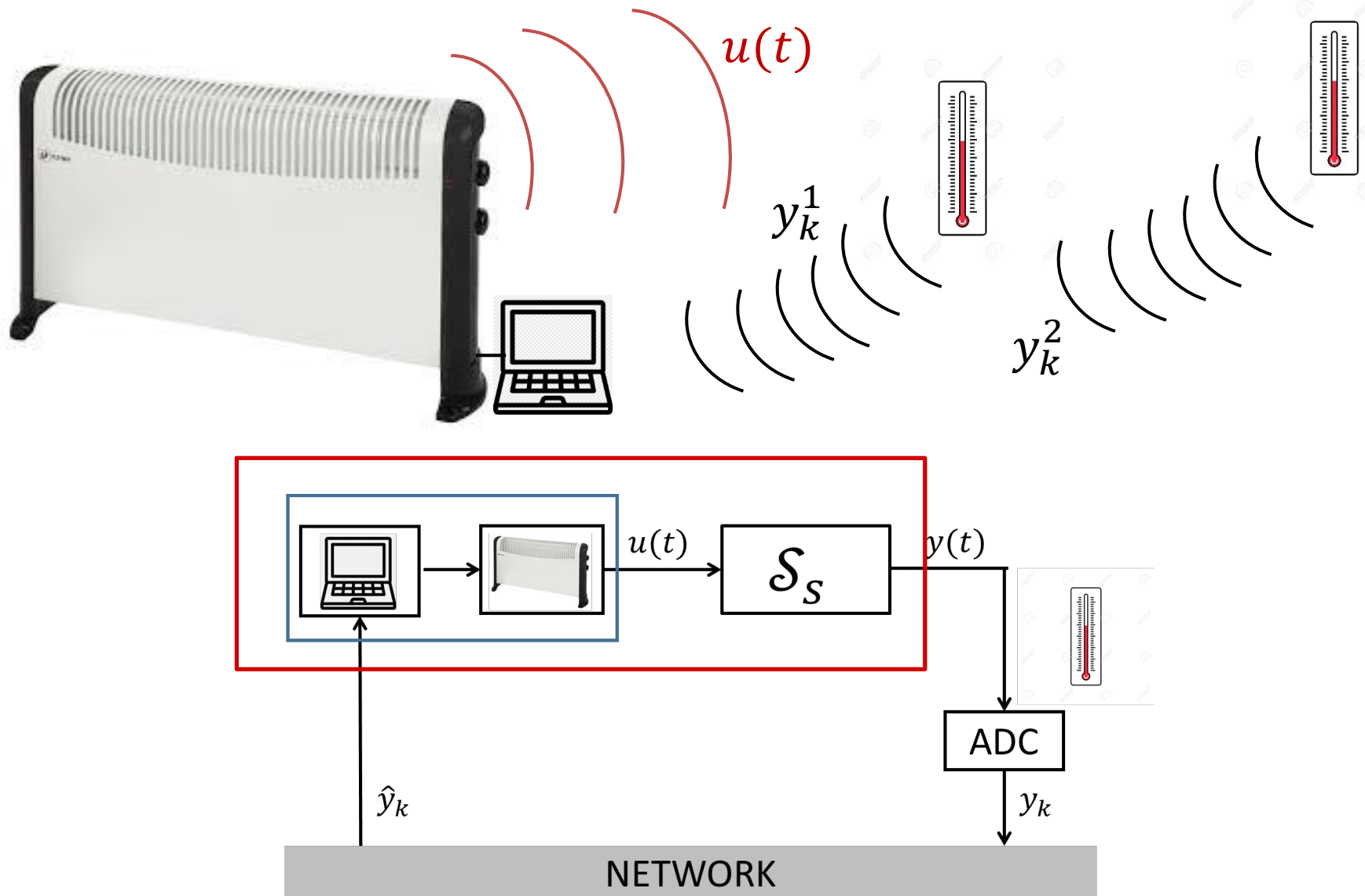
$$u_k = S(x_k) = \begin{cases} 1 & x_k < -\frac{1}{2} \\ 0 & -\frac{1}{2} \leq x_k < 0 \\ 0 & 0 \leq x_k < \frac{1}{2} \\ 1 & x_k \geq \frac{1}{2} \end{cases}$$

2. DELAYS

A. ASSUMPTIONS

1) Configuration (see slider below)

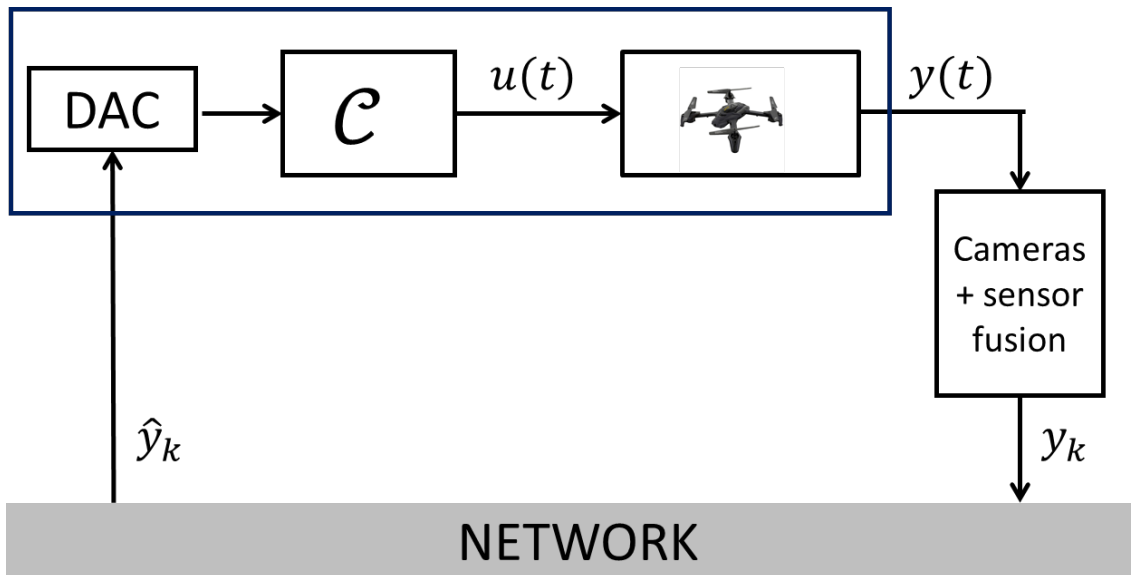
Reference configuration - examples



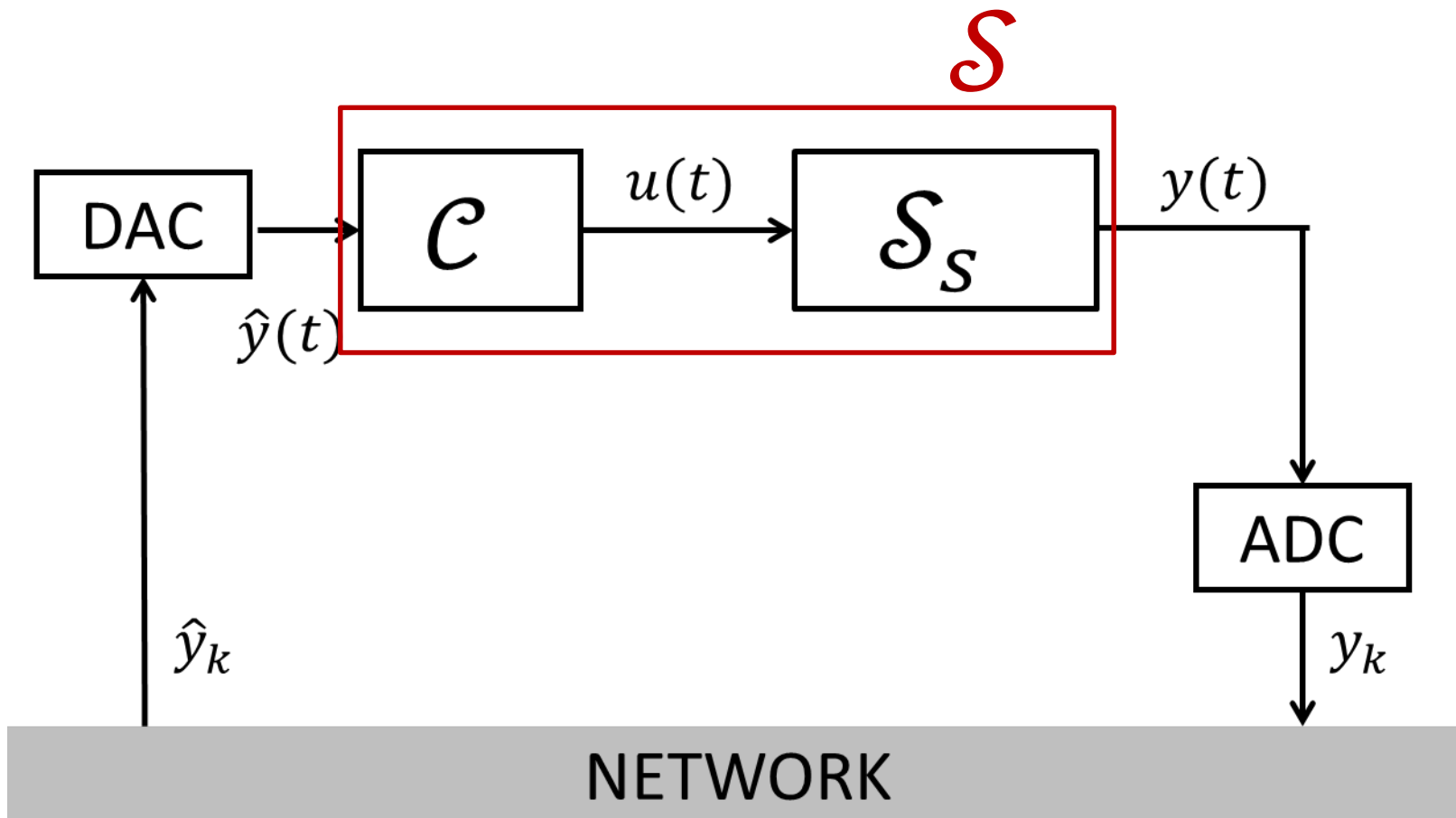
Reference configuration - examples

Flying arena

- Cameras (e.g., IR) sense distances
- Local transmission for sensor fusion (wired)
- Position y_k sent to the quadrotor on-board control station through wireless network

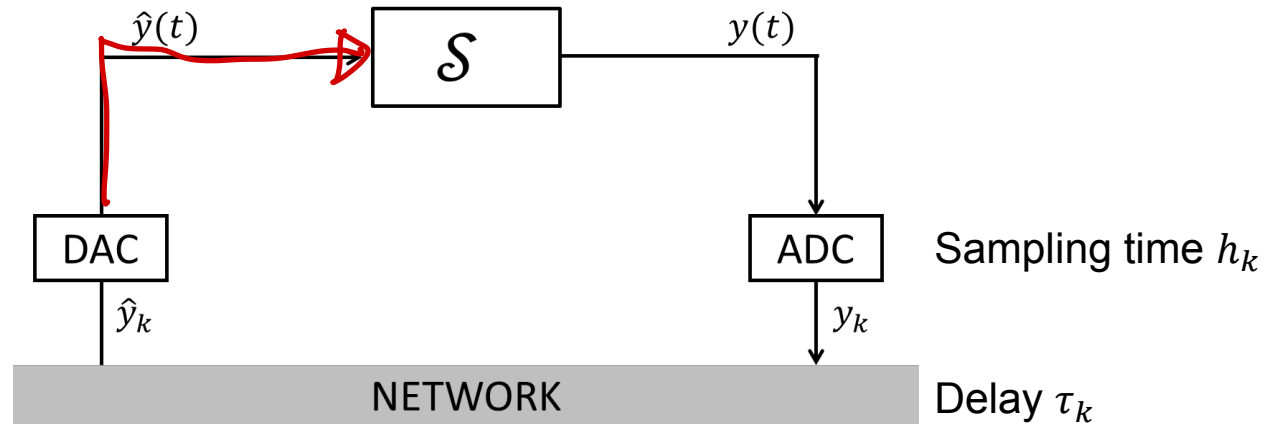


Reference configuration



Reference configuration

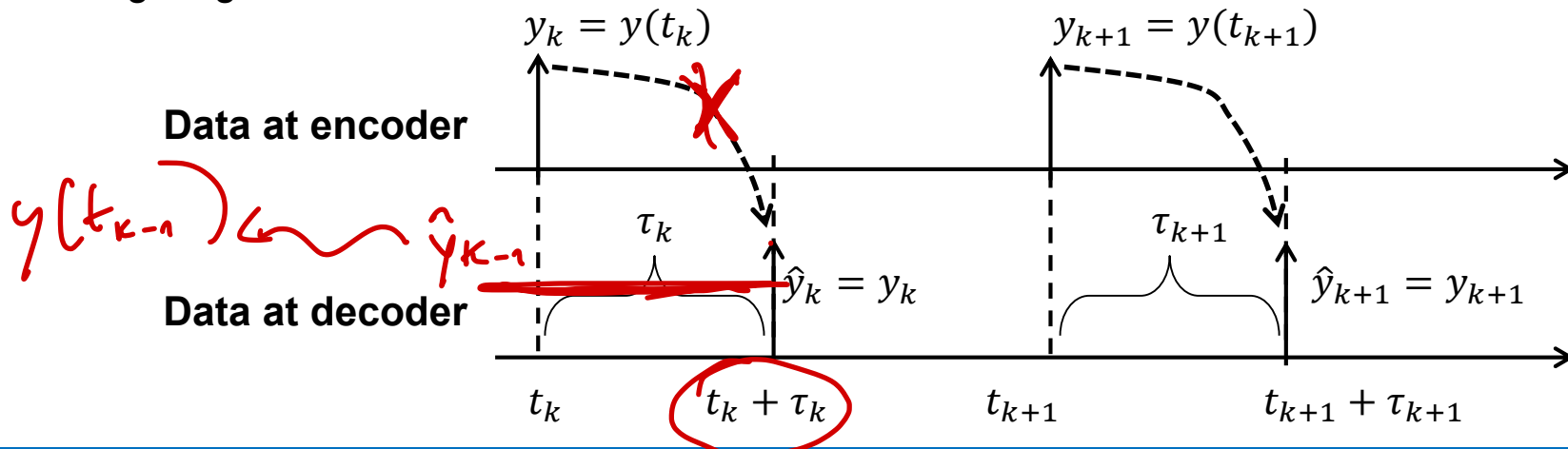
In this lecture we consider the general NCS scheme



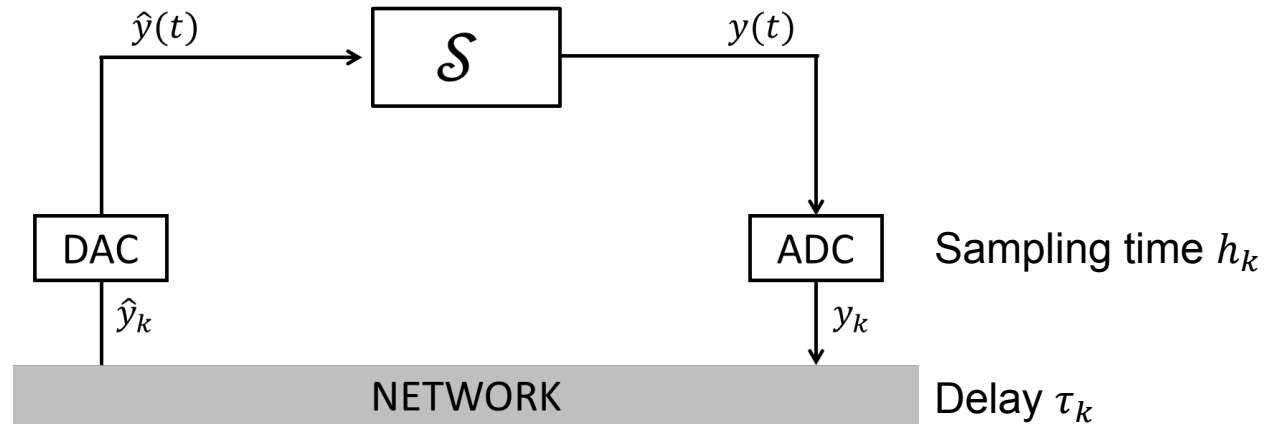
where the dynamics of $\mathcal{S}$ is

$$\begin{cases} \dot{x}(t) = Ax(t) + B\hat{y}(t) \\ y(t) = Cx(t) \end{cases}$$

Timing diagram



Reference configuration

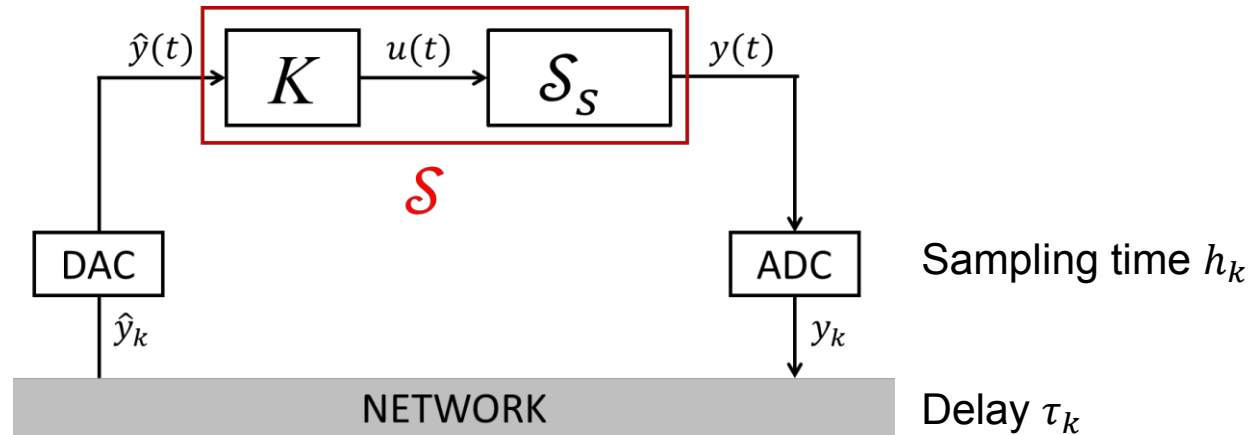


This scheme is very general/generalizable!

In fact we can describe, through the above scheme (and by proper reformulations), a number of different notable configurations (control systems), e.g.,

- Static and dynamic output-feedback controller co-located with the actuator
- Static and dynamic output-feedback controller co-located with the sensor
- Remote state-feedback controller
- ...

The case of static output-feedback control co-located with the actuator



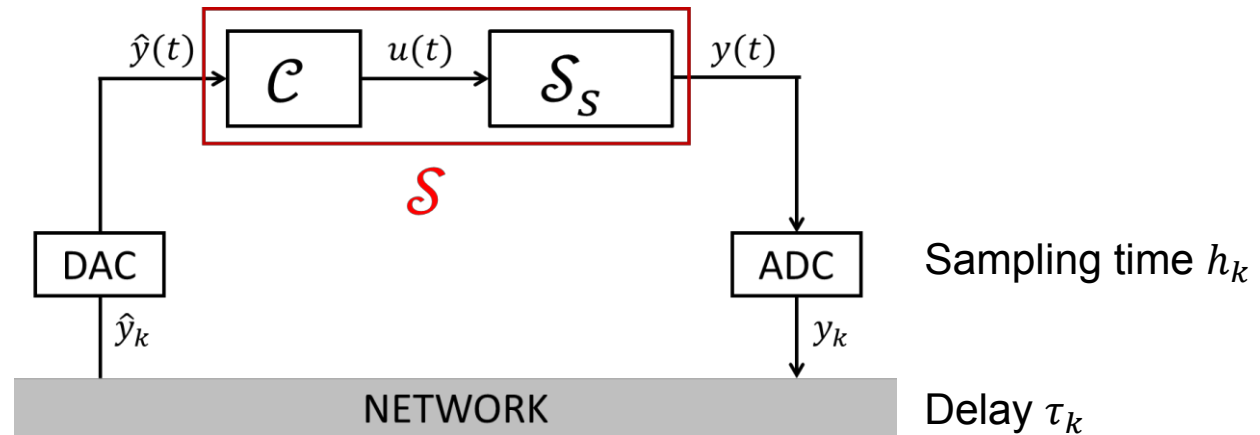
The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. Renaming $x(t) = x_S(t)$, $A = A_S$, $B = B_S K$, $C = C_S$, we retrieve the scheme of the general configuration, i.e.,

$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

The case of dynamic output-feedback control co-located with the actuator



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and the dynamics of the dynamic output-feedback controller $\mathcal{C}$ is

$$\begin{cases} \dot{x}_C(t) &= A_C x_C(t) + B_C \hat{y}(t) \\ u(t) &= C_C x_C(t) + D_C \hat{y}(t) \end{cases}$$

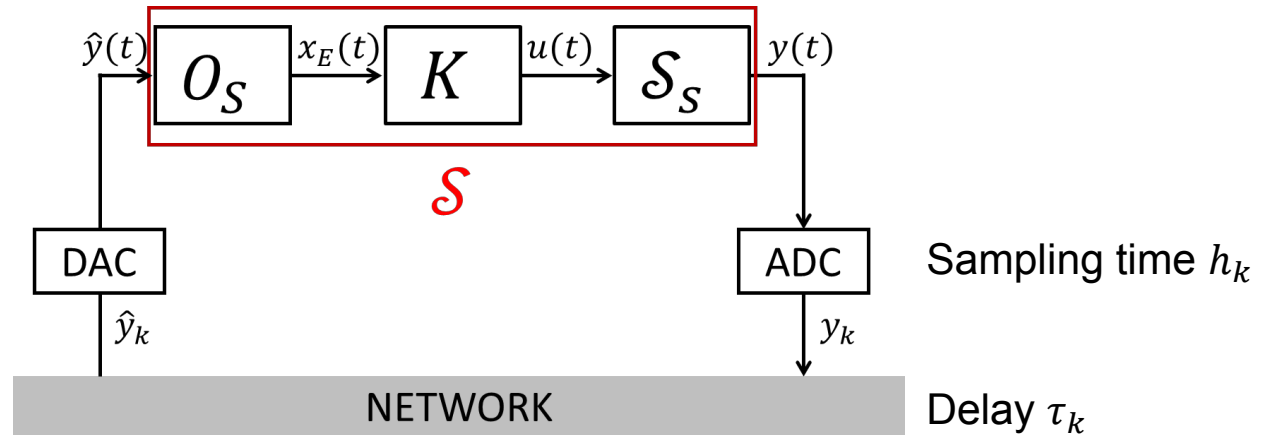
Overall, we write

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S (C_C x_C(t) + D_C \hat{y}(t)) \\ \dot{x}_C(t) &= A_C x_C(t) + B_C \hat{y}(t) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_C(t) \end{bmatrix}$, $A = \begin{bmatrix} A_S & B_S C_C \\ 0 & A_C \end{bmatrix}$, $B = \begin{bmatrix} B_S D_C \\ B_C \end{bmatrix}$, $C = [C_S \quad 0]$, we obtain

$$\begin{cases} \dot{x}(t) &= Ax(t) + B \hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

The case of observer + static state-feedback control co-located with the actuator



The dynamics of S_S is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

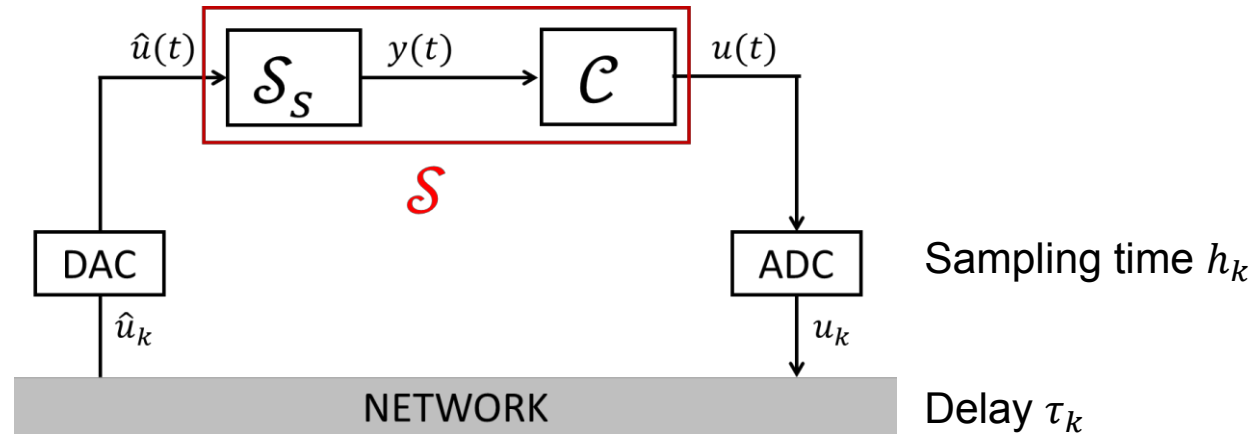
the dynamics of the observer O_S is, e.g., $\dot{x}_E(t) = A_S x_E(t) + B_S u(t) + L_E(\hat{y}(t) - C_S x_E(t))$, and the control action is set to $u(t) = K x_E(t)$. Overall, we write

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S K x_E(t) \\ \dot{x}_E(t) &= A_S x_E(t) + B_S K x_E(t) + L_E(\hat{y}(t) - C_S x_E(t)) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_E(t) \end{bmatrix}$, $A = \begin{bmatrix} A_S & B_S K \\ 0 & A_S + B_S K - L_E C_S \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ L_E \end{bmatrix}$, $C = [C_S \quad 0]$, we obtain

$$\begin{cases} \dot{x}(t) &= A x(t) + B \hat{y}(t) \\ y(t) &= C x(t) \end{cases}$$

The case of output-feedback control co-located with the sensor



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S \hat{u}(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and the dynamics of the dynamic output-feedback controller $\mathcal{C}$ is

$$\begin{cases} \dot{x}_C(t) &= A_C x_C(t) + B_C y(t) \\ u(t) &= C_C x_C(t) + D_C y(t) \end{cases}$$

Overall, we write

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S \hat{u}(t) \\ \dot{x}_C(t) &= A_C x_C(t) + B_C C_S x_S(t) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_C(t) \end{bmatrix}^T$, $A = \begin{bmatrix} A_S & 0 \\ B_C C_S & A_C \end{bmatrix}$, $B = \begin{bmatrix} B_S \\ 0 \end{bmatrix}$, $C = [D_C C_S \quad C_C]$, we obtain

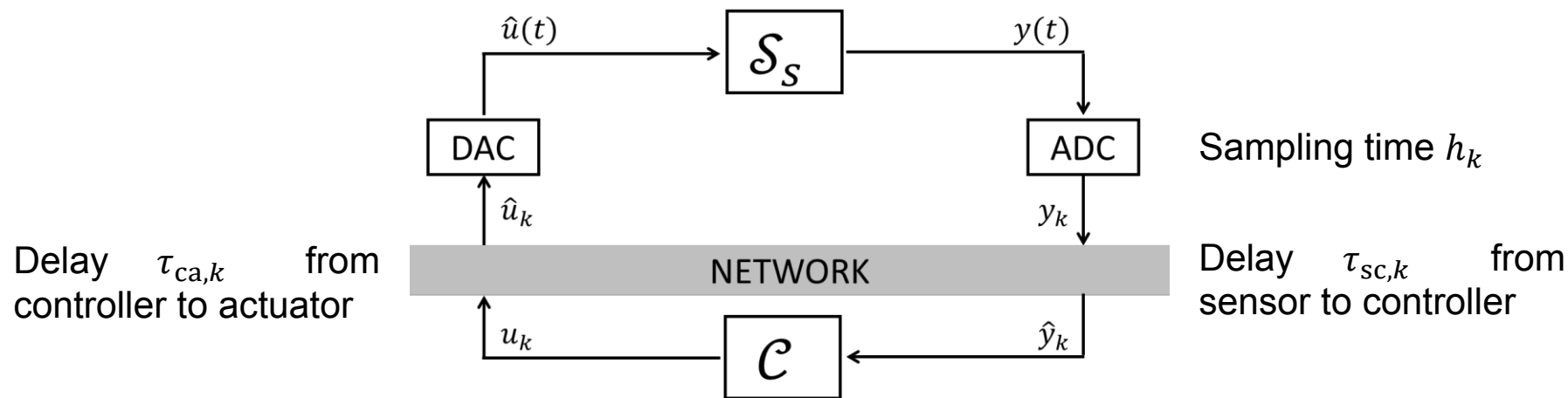
$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{u}(t) \\ u(t) &= Cx(t) \end{cases} \Rightarrow$$

Original scheme where $y(t) \rightarrow u(t)$ and $\hat{y}(t) \rightarrow \hat{u}(t)$

Remarks:

- Setting $C_C = 0$, $D_C = K$ this is the static output-feedback scheme
- With suitable definition of C_C and D_C also the scheme with the observer is considered

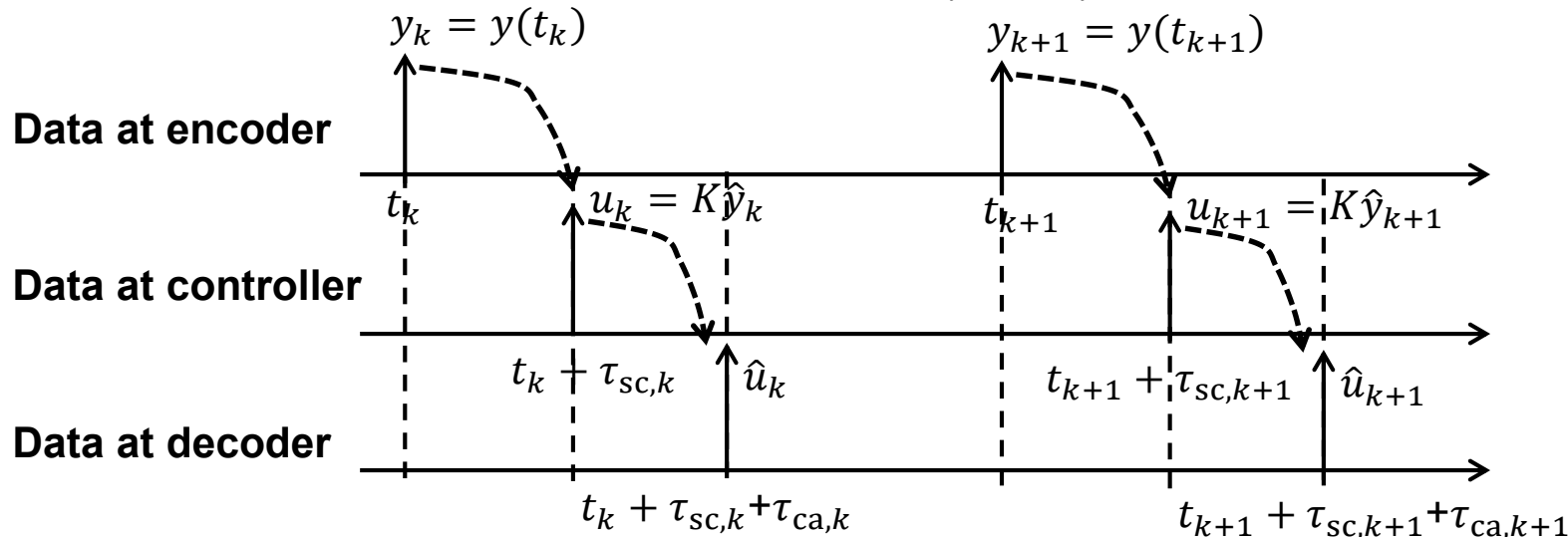
The case of **output-feedback** **remote** control



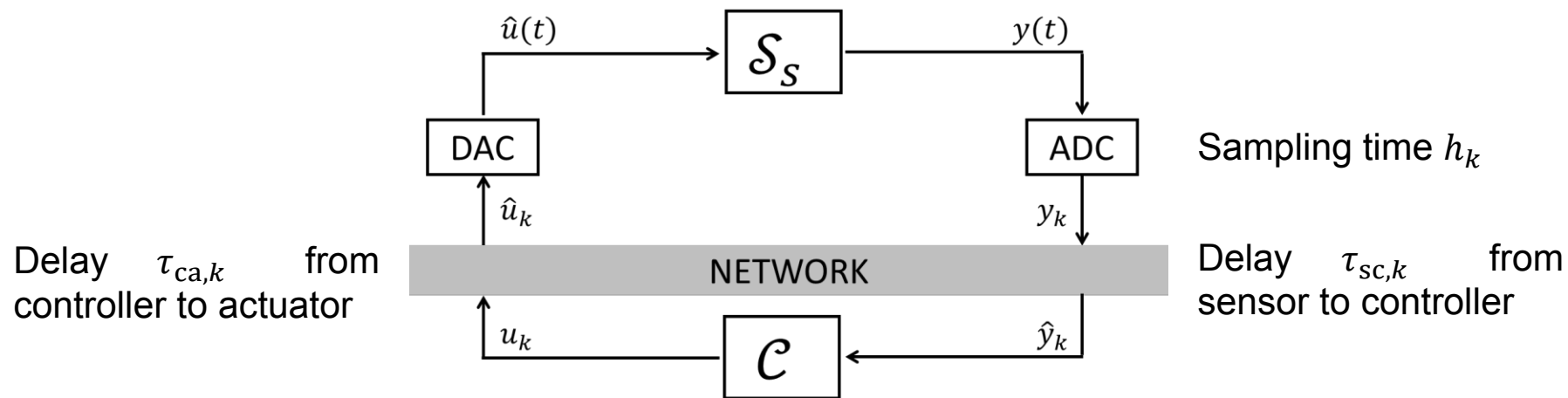
The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}$$

An output-feedback static controller is used, i.e., $u_k = K \hat{y}_k$.



The case of output-feedback static remote control

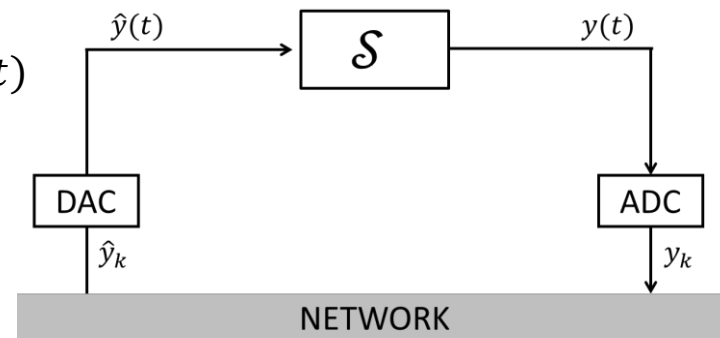


Renaming $x(t) = x_S(t)$, $A = A_S$, $B = B_S K$, $C = C_S$, this scheme is equivalent to the one of the general configuration, i.e.,

$$\begin{cases} \dot{x}(t) = Ax(t) + B\hat{y}(t) \\ y(t) = Cx(t) \end{cases}$$

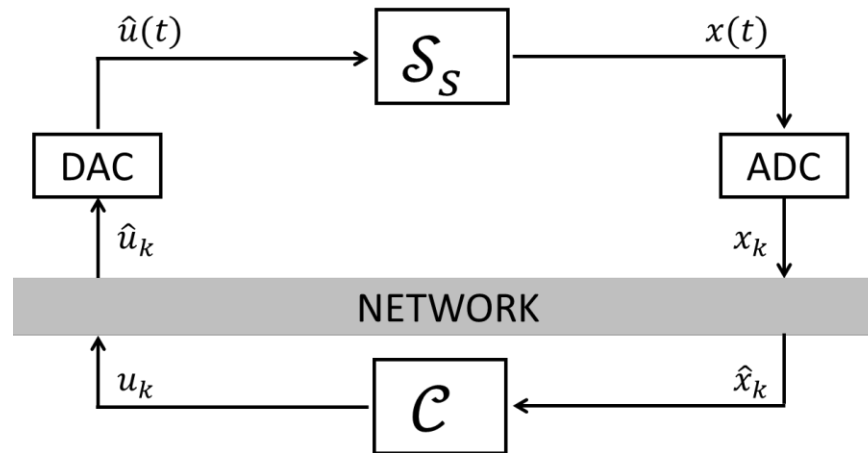
provided that the (unique) delay is set to

$$\tau_k = \tau_{sc,k} + \tau_{ca,k}$$

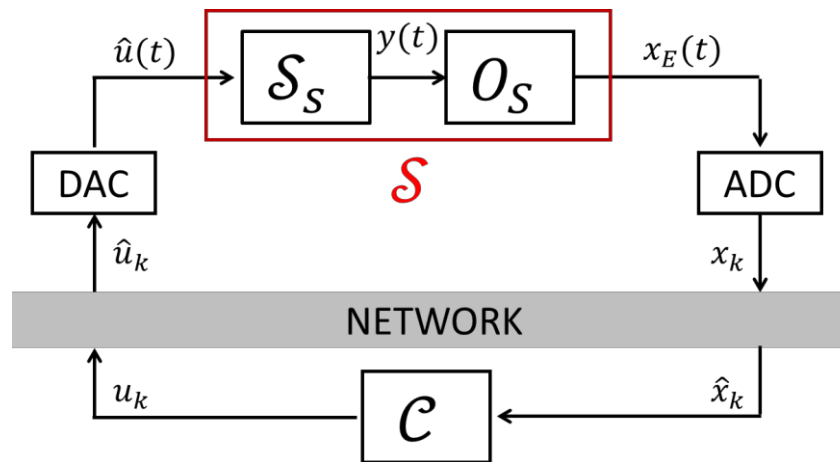


The case of **general-type** remote control

If $C_S = I$ a static state-feedback controller is obtained



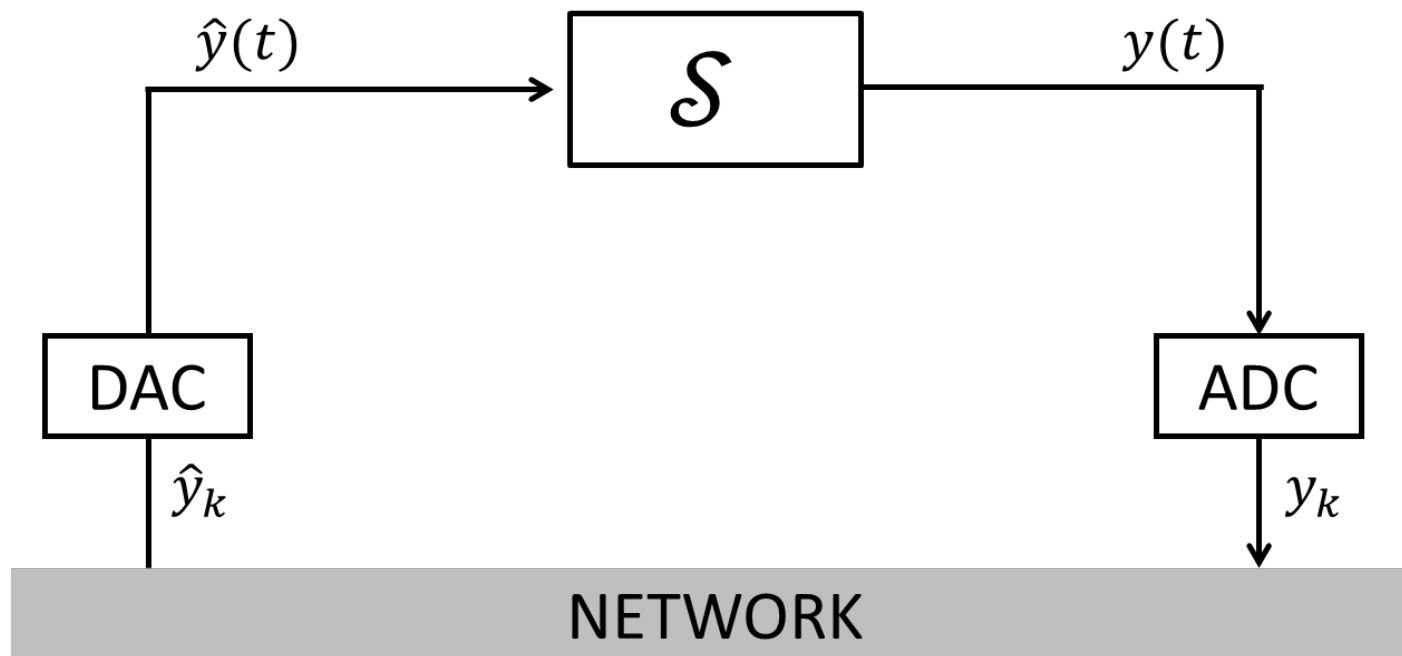
If $\mathcal{S}$ includes both system and state observer, also the case of dynamic output-feedback control can be considered



Reference configuration

In conclusion, the scheme

$$\mathcal{S}: \begin{cases} \dot{x}(t) = Ax(t) + B\hat{y}(t) \\ y(t) = Cx(t) \end{cases}$$

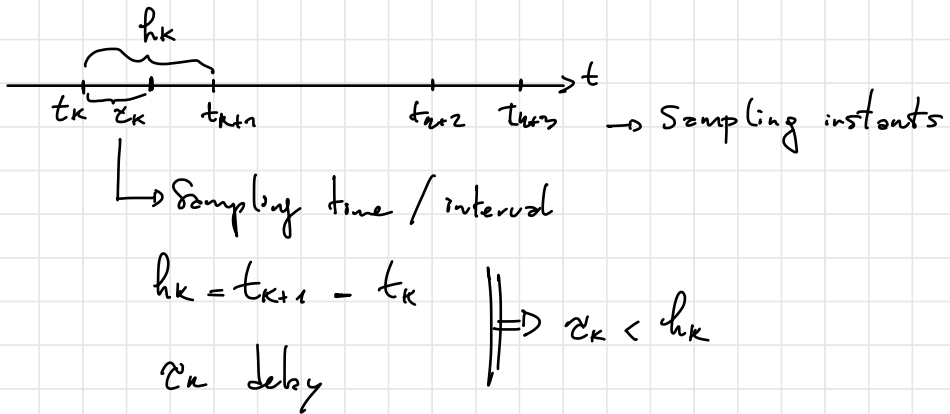


is a suitable abstraction for an extremely wide variety of control systems, among which the ones with a remote controller

2) Network channel

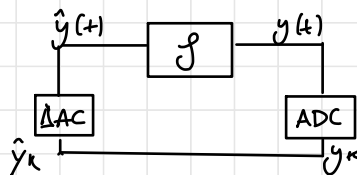
- Packets are enough "large" to consider the transmitted data as "real" $\rightarrow$ no approximation
- Network issues :
 - DELAYS
 - PACKET DROPOUTS

3) Sampling



B. MODEL OF THE NCS (Networked control system)

- IF THERE IS NO DELAY, THE (DIGITAL) CONTROL SYSTEM IS :



$$\begin{cases} \dot{x}(t) = Ax(t) + B\hat{y}(t) \\ y(t) = Cx(t) \end{cases} \xrightarrow[\substack{\text{DISCRETIZATION} \\ \text{SAMPLING TIME } h}]{\text{}} \begin{cases} x_{k+1} = Fx_k + G\hat{y}_k \\ y_k = Hx_k \end{cases}$$

CLOSED LOOP: $\hat{y}_k = y_k = H x_k$

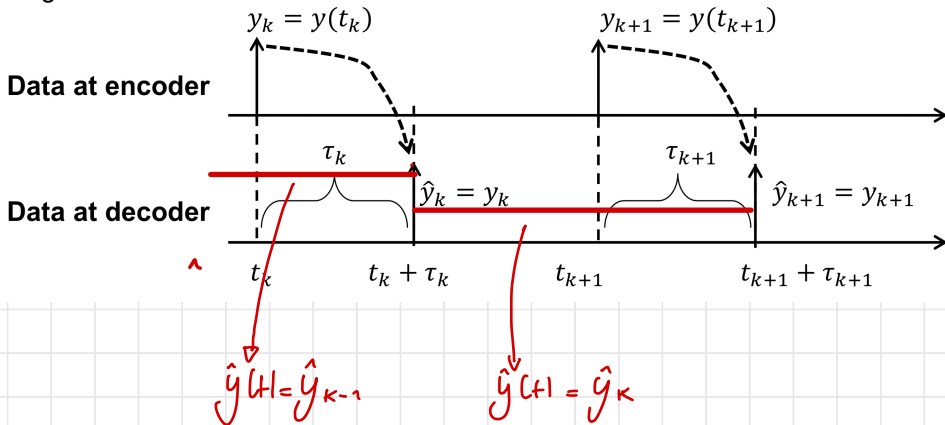
$$\Rightarrow x_{k+1} = F x_k + G H x_k$$

$$= (F + GH) x_k$$

• IF WE HAVE DELAYS

$$j: \begin{cases} \dot{x}(t) = A x(t) + B \hat{y}(t) \\ y(t) = C x(t) \end{cases}$$

Timing diagram



$$\hat{y}(t) = \begin{cases} \hat{y}_{k-1} & t \in [t_k, t_k + \tau_k) \\ \hat{y}_k & t \in [t_k + \tau_k, t_{k+1}) \end{cases}$$

We use the Lagrange equation:

$$x(t_{k+1}) = e^{A(t_{k+1} - t_k)} x(t_k) + \int_{t_k}^{t_{k+1}} e^{A(t_{k+1} - \nu)} B \hat{y}(\nu) d\nu$$

$\underbrace{x(t_{k+1})}_{x_{k+1}} = e^{\underbrace{A(t_{k+1} - t_k)}_{h_k}} \underbrace{x(t_k)}_{x_k} + \underbrace{\int_{t_k}^{t_{k+1}} e^{A(t_{k+1} - \nu)} B \hat{y}(\nu) d\nu}_{(i)} + \underbrace{\int_{t_k + \tau_k}^{t_{k+1}} e^{A(t_{k+1} - \nu)} B \hat{y}(\nu) d\nu}_{(ii)}$

$$(i) = \int_{t_k}^{t_k + \tau_k} e^{A(t_{k+1} - v)} B \hat{y}_{k-1} dv = \int_{t_k}^{t_k + \tau_k} e^{A(t_{k+1} - v)} dv B \hat{y}_{k-1}$$

$$= \int_{t_k}^{t_k + \tau_k} e^{A(t_{k+1} - (t_k + \tau_k) + (t_k + \tau_k) - v)} dv$$

$$= \int_{t_k}^{t_k + \tau_k} e^{A(t_{k+1} - t_k - \tau_k)} e^{A(t_k + \tau_k - v)} dv$$

$$= e^{A(t_{k+1} - t_k - \tau_k)} \int_{t_k}^{t_k + \tau_k} e^{A(t_k + \tau_k - v)} dv$$

$$t_k + h_k - t_k - \tau_k = h_k - \tau_k (> 0)$$

$$\int_{t_k}^{t_k + \tau_k} e^{A(t_k + \tau_k - v)} dv$$

$$\text{WHEN } v = t_k \Rightarrow t_k + \tau_k - v = \tau_k$$

$$\text{WHEN } v = t_k + \tau_k \Rightarrow t_k + \tau_k - v = 0$$

$\Downarrow$

$$= \int_0^{\tau_k} e^{A v} dv = \Gamma(\tau_k)$$

$$= e^{A(h_k - \tau_k)} \Gamma(\tau_k) B \hat{y}_{k-1}$$

$$(ii) = \int_{t_k + \tau_k}^{t_{k+1}} e^{A(t_{k+1} - v)} B \hat{y}(v) dv = \int_{t_k + \tau_k}^{t_k + h_k} e^{A(t_k + h_k - v)} dv B \hat{y}_k$$

$$\hat{y}_k$$

$$\text{WHEN } v = t_k + \tau_k \Rightarrow t_k + h_k - v = h_k - \tau_k$$

$$\text{WHEN } v = t_k + h_k \Rightarrow t_k + h_k - v = 0$$

$$= \int_0^{h_k - \tau_k} e^{A v} dv = \Gamma(h_k - \tau_k)$$

$$= \Gamma(h_k - \tau_k) B \hat{y}_k$$

OVERALL :

$$\Rightarrow x_{k+1} = e^{A h_k} x_k + e^{A(h_k - \tau_k)} \Gamma(\epsilon_k) B \hat{y}_{k-1} + \Gamma(h_k - \tau_k) B \hat{y}_k$$

$$y_k = C x_k$$

WE DEFINE WITH z_k THE STATE OF THE NCS

$$z_k = \begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix} \Rightarrow z_{k+1} = \begin{cases} x_{k+1} = (e^{A h_k} + \Gamma(h_k - \tau_k) B C) x_k + e^{A(h_k - \tau_k)} \Gamma(\epsilon_k) B \hat{y}_{k-1} \\ \hat{y}_k = C x_k \end{cases}$$

$$\downarrow \Rightarrow z_{k+1} = \underbrace{\begin{bmatrix} e^{A h_k} + \Gamma(h_k - \tau_k) B C & e^{A(h_k - \tau_k)} \Gamma(\epsilon_k) B \\ C & 0 \end{bmatrix}}_{\Phi(h_k, \tau_k)} \underbrace{\begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix}}_{z_k}$$

$\Rightarrow$ DYNAMICS OF THE NCS :

$$z_{k+1} = \Phi(h_k, \tau_k) z_k \Rightarrow \text{NOT LTI}$$

$\hookrightarrow$ Since h_k, τ_k are time-varying

$\Phi(h_k, \tau_k)$ is time varying

$\rightarrow$ • LTV

LINEAR TIME-VARYING SYSTEM

• LPV

LINEAR PARAMETER-VARYING SYSTEM

C. STABILITY OF THE NCS

1) The case of constant h_k and τ_k

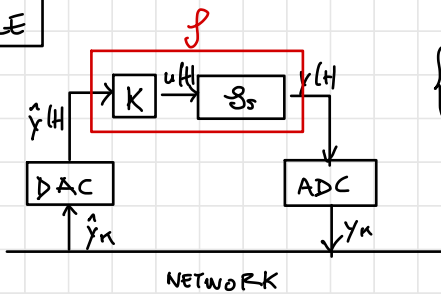
Assumption 1: $h_k = h$, $\tau_k = \tau \forall k$

In this case, $\Phi(h_k, \tau_k) = \Phi(h, \tau)$

$$x_{k+1} = \Phi(h, \tau) x_k \quad \text{LTI system}$$

Theorem: Under Assumption 1, the NCS is asymptotically (exponentially) stable if and only if $\Phi(h, \tau)$ is Schur stable.

EXAMPLE



$$\begin{cases} \dot{x}_s = A_s x_s + B_s u = A_s x_s + B_s K \hat{y} \\ y = C_s x_s \end{cases}$$

$$u = K \hat{y}$$

$$x_s \rightarrow x$$

$$\begin{cases} \dot{x} = A_s x + \boxed{B_s K} \hat{y} \\ y = C_s x \end{cases} \Rightarrow \begin{aligned} A &= A_s \\ B &= B_s K \\ C &= C_s \end{aligned}$$

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_s h} + \Gamma(h, \tau) B_s K C_s & e^{A_s(h-\tau)} \Gamma(\tau) B_s K \\ C_s & 0 \end{bmatrix}$$

CASE 1

$\dot{X}_3 = u$ integrator

$$\downarrow$$

$$n=1, \quad A_s=0, \quad B_s=1, \quad C_s=1$$

$$\Rightarrow \begin{aligned} & \cdot e^{A_s h} = e^0 = 1 \\ & \cdot \Gamma(\mu) = \int_0^\mu e^{A_s v} dv = \int_0^\mu e^0 dv = \mu \\ & \cdot \Gamma(h-z) = h-z \\ & \cdot \Gamma(z) = z \end{aligned}$$

$$\Rightarrow \underline{I}(h, z) = \begin{bmatrix} 1 + (h-z)k & Kz \\ 1 & 0 \end{bmatrix}$$

$$\circ \text{ IF } z=0 \Rightarrow \underline{I}(h, z) = \begin{bmatrix} 1 + hk & 0 \\ 1 & 0 \end{bmatrix}$$

CONDITION FOR STABILITY :

$$\begin{aligned} & |1 + hk| < 1 \Rightarrow \text{THIS CONDITION IS THE} \\ & \downarrow \text{ONE THAT WE OBTAIN} \\ & -1 < 1 + hk < 1 \text{ IN CASE OF DIGITAL} \\ & \downarrow \text{CONTROL} \end{aligned}$$

$$\begin{cases} K < 0 \\ Kh > -2 \end{cases}$$

$$\Rightarrow \boxed{-\frac{2}{h} < K < 0}$$

$$\circ \text{ IF } 0 < z < h$$

$$\underline{I}(h, z) = \begin{bmatrix} 1 + (h-z)k & Kz \\ 1 & 0 \end{bmatrix}$$

$$\begin{aligned}
 p(\lambda) &= \det(\lambda I - \mathbb{E}(h, \tau)) \\
 &= \det \begin{bmatrix} \lambda - (1 + (h - \tau)k) & -k\tau \\ -1 & \lambda \end{bmatrix} \\
 &= \lambda^2 - \underbrace{(1 + (h - \tau)k)}_a \lambda - \underbrace{k\tau}_b
 \end{aligned}$$

FROM THE JURY CRITERION,
 $\mathbb{E}(h, \tau)$ IS SCHUR IFF:

$$\begin{aligned}
 -1 < b < 1 \\
 -(1+b) < -a < (1+b)
 \end{aligned}$$

$$\begin{cases}
 -1 < k\tau < 1 & (i) \\
 1 + (h - \tau)k < 1 - k\tau & (ii) \\
 1 + (h - \tau)k > -1 + k\tau & (iii)
 \end{cases}$$

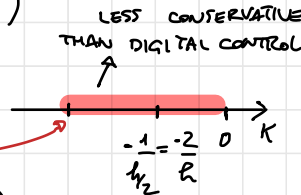
$$k_h < 0 \Rightarrow k < 0$$

$$(h - 2\tau)k > -2 \iff \left(\frac{h}{2} - \tau\right)k > -1$$

• IF $\tau < \frac{h}{2}$ (small delay)

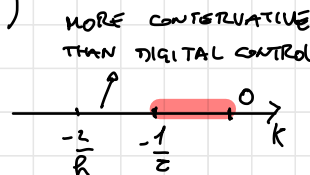
$$\begin{cases}
 k < 0 \\
 k\tau > -1 \\
 \left(\frac{h}{2} - \tau\right)k > -1
 \end{cases}$$

$$\max\left(-\frac{1}{\tau}, \frac{-1}{\frac{h}{2} - \tau}\right) < k < 0$$



• IF $\tau > \frac{h}{2}$ (large delay)

$$-\frac{1}{\tau} < k < 0$$



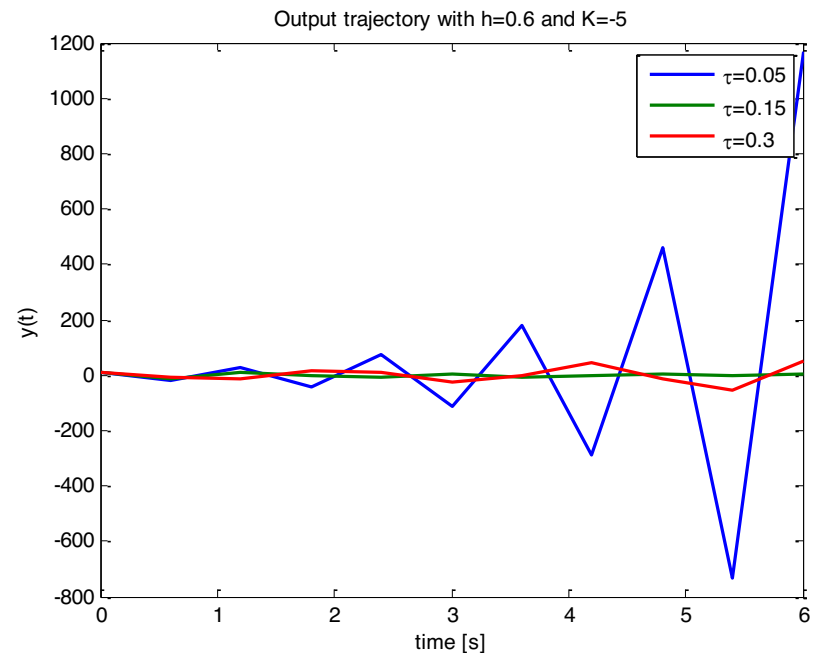
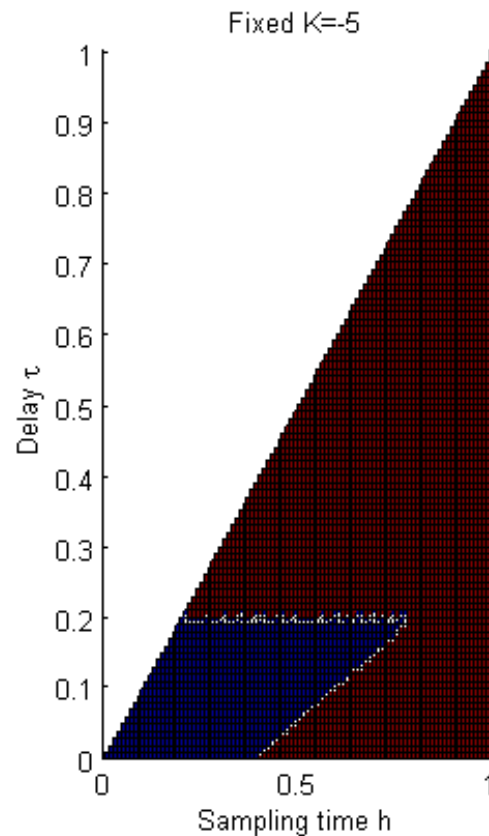
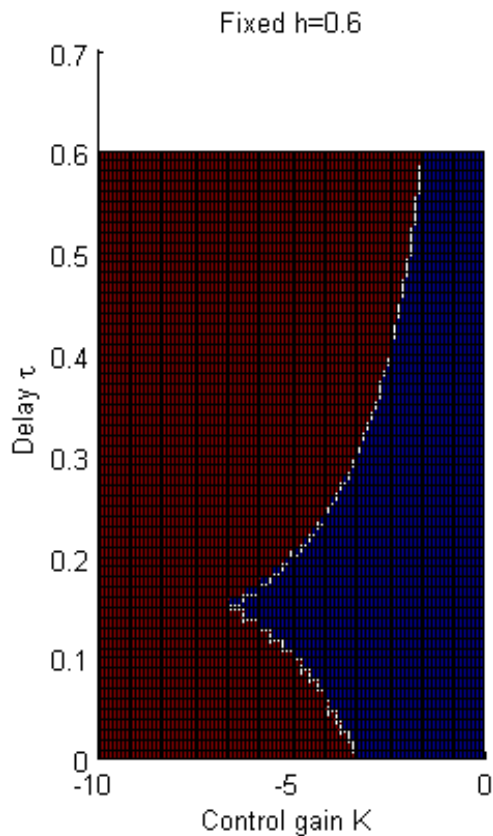
Example 1

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= u(t) \\ y(t) &= x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} 1 + K(h - \tau) & K\tau \\ 1 & 0 \end{bmatrix}$$



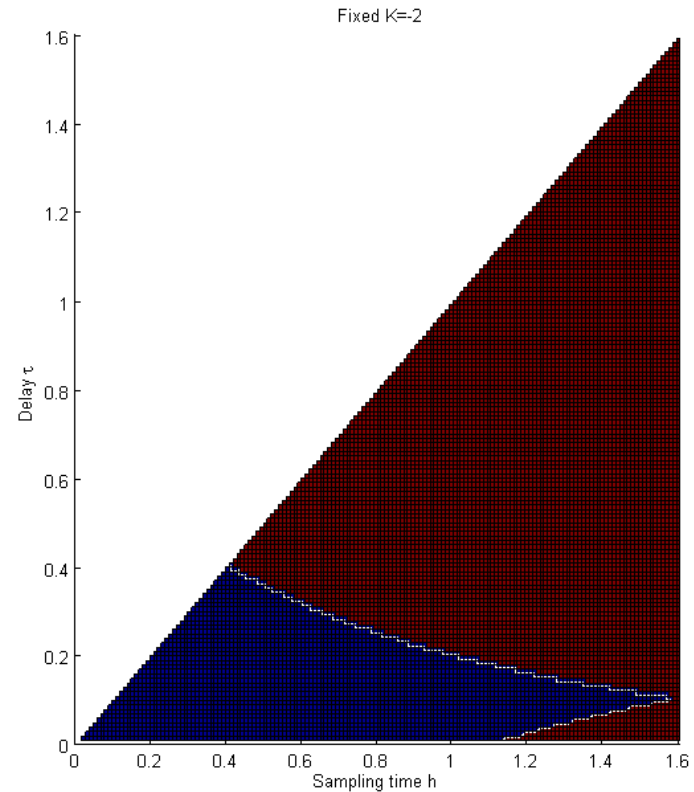
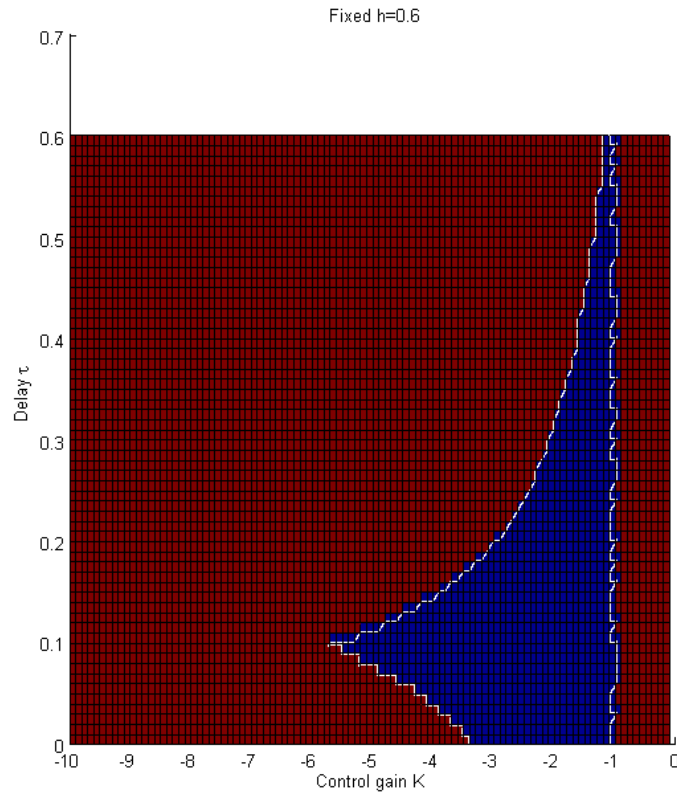
Example 2

The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^h + K(e^{h-\tau} - 1) & Ke^{h-\tau}(e^\tau - 1) \\ 1 & 0 \end{bmatrix}$$



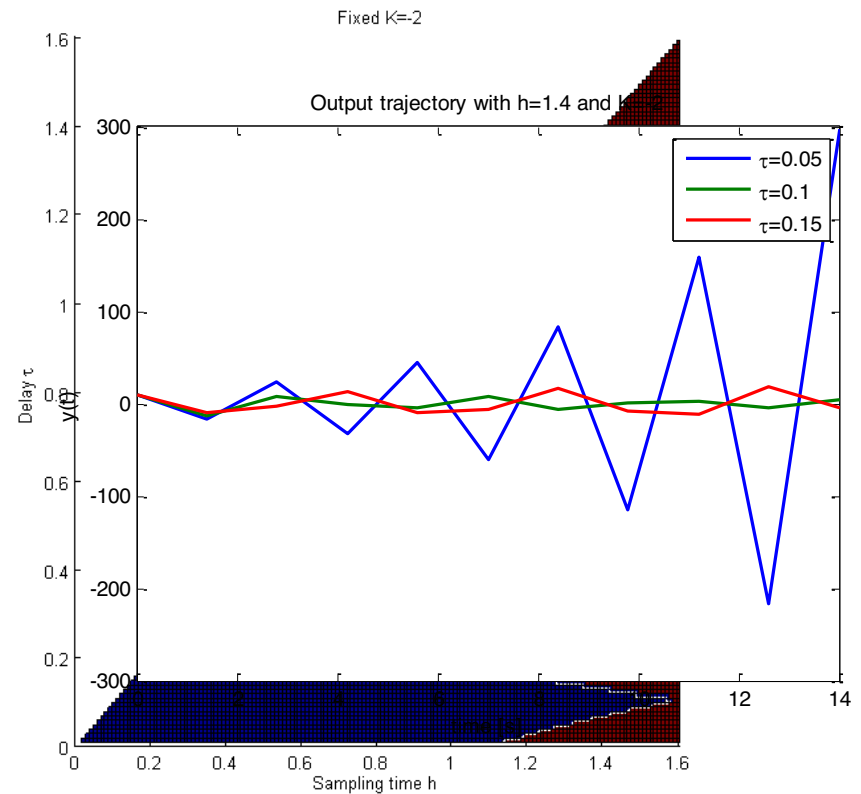
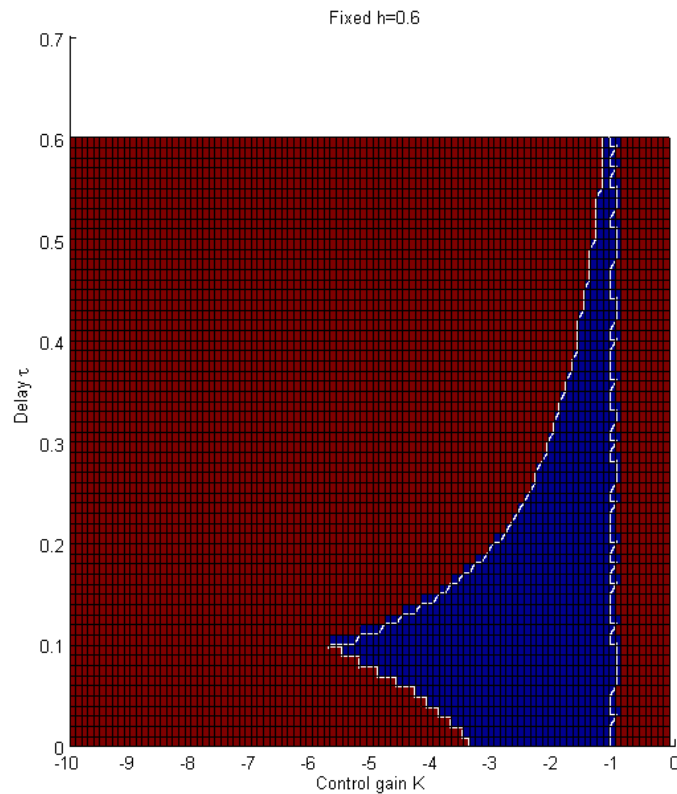
Example 2

The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^h + K(e^{h-\tau} - 1) & Ke^{h-\tau}(e^\tau - 1) \\ 1 & 0 \end{bmatrix}$$



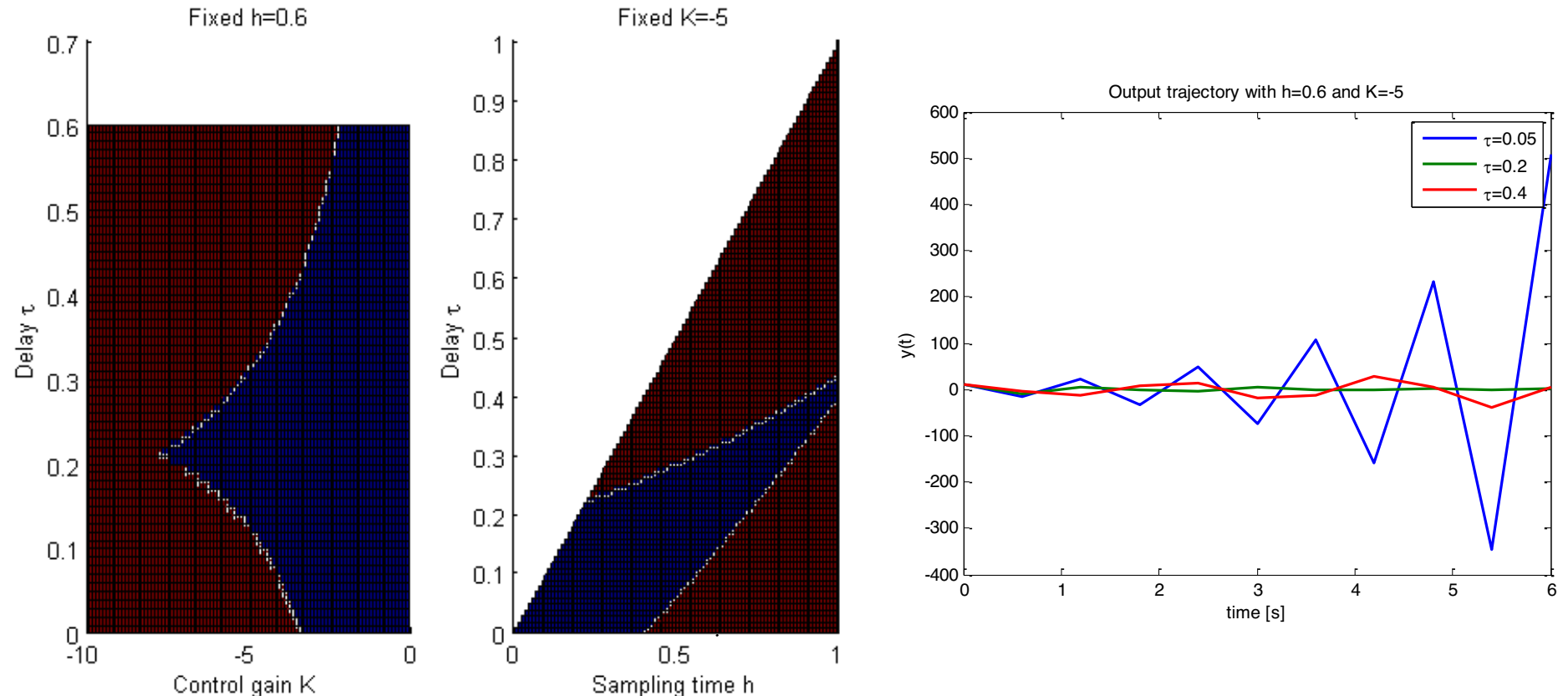
Example 3

The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= -x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^{-h} + K(1 - e^{-(h-\tau)}) & Ke^{-(h-\tau)}(1 - e^{-\tau}) \\ 1 & 0 \end{bmatrix}$$



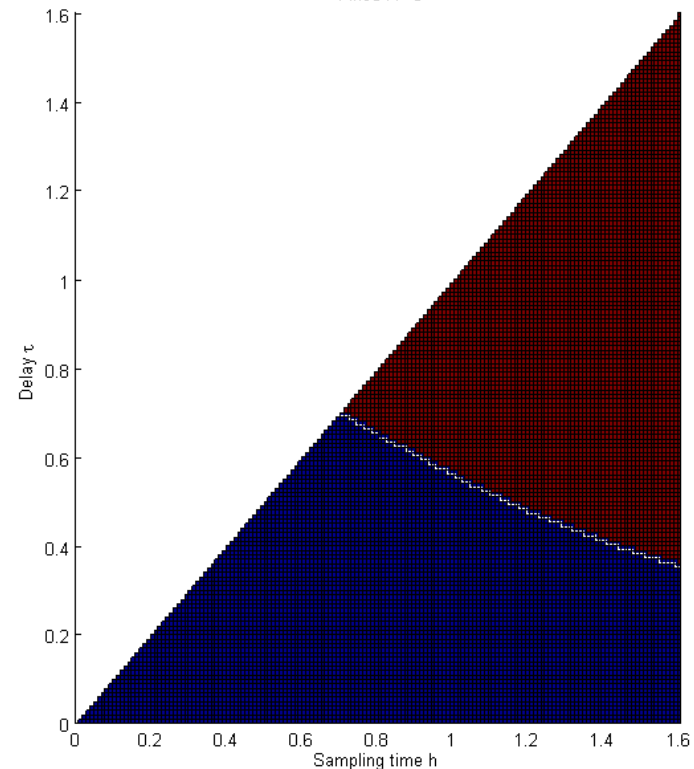
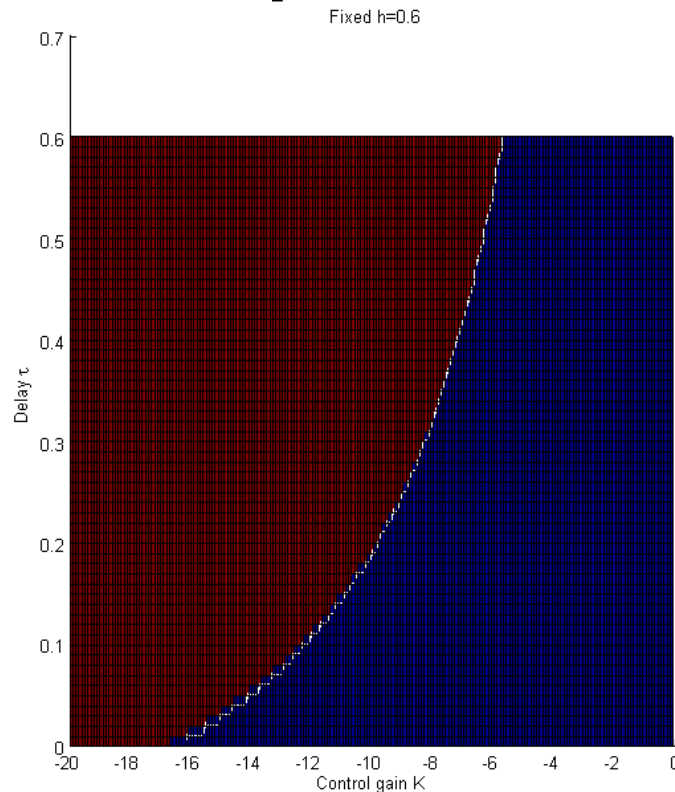
Example 4

The dynamics of the 2x2 system $\mathcal{S}$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}, A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = \begin{bmatrix} -0.1 & 1 \end{bmatrix}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_S h} + A_S^{-1}(e^{A_S(h-\tau)} - I)B_S K C_S & e^{A_S(h-\tau)} A_S^{-1}(e^{A_S \tau} - I)B_S K \\ C_S & 0 \end{bmatrix}$$



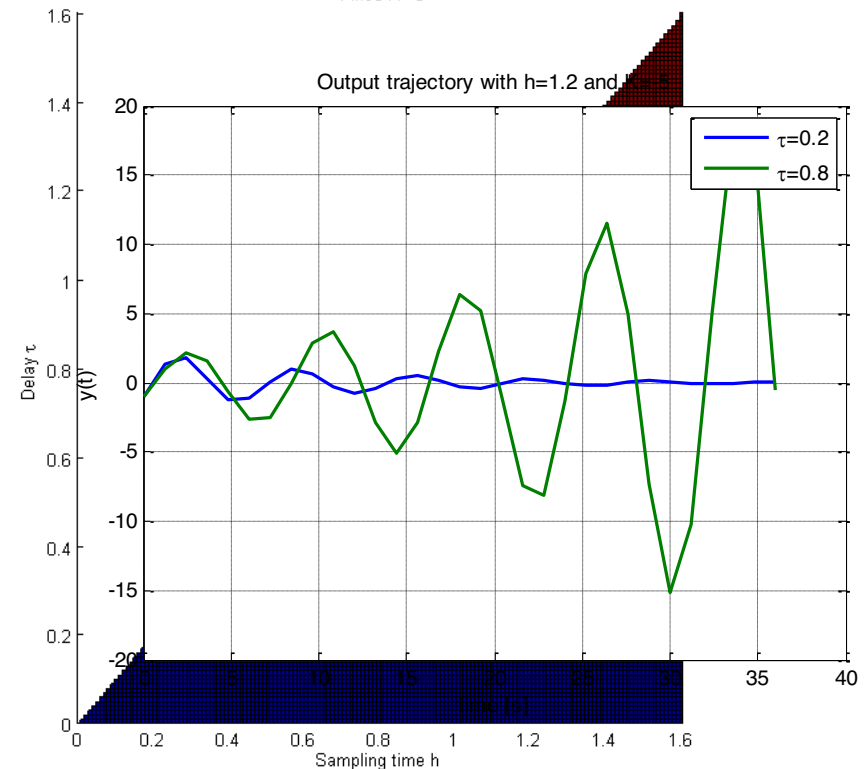
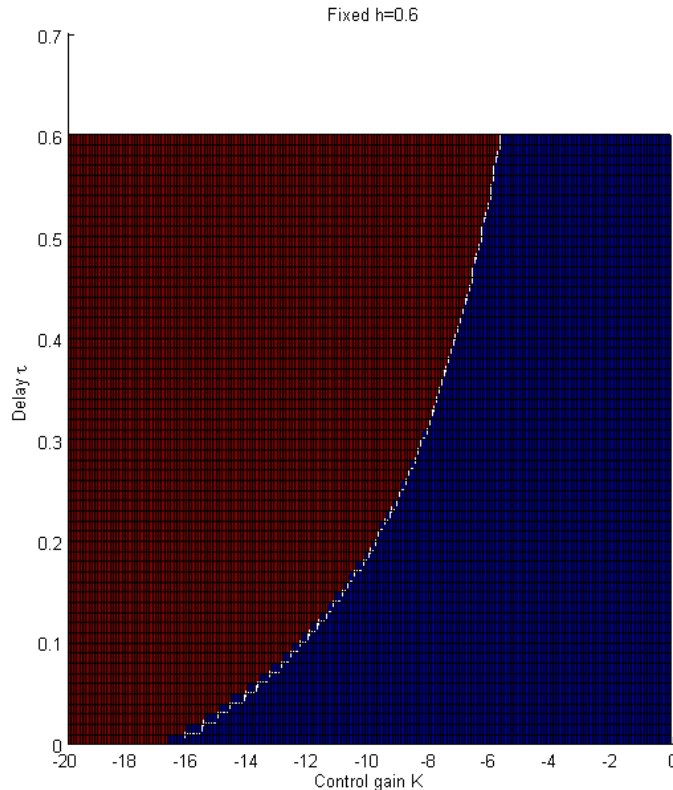
Example 4

The dynamics of the 2x2 system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) = A_s x_s(t) + B_s u(t) \\ y(t) = C_s x_s(t) \end{cases}, A_s = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_s = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_s = \begin{bmatrix} -0.1 & 1 \end{bmatrix}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_s h} + A_s^{-1}(e^{A_s(h-\tau)} - I)B_s K C_s & e^{A_s(h-\tau)} A_s^{-1}(e^{A_s \tau} - I)B_s K \\ C_s & 0 \end{bmatrix}$$

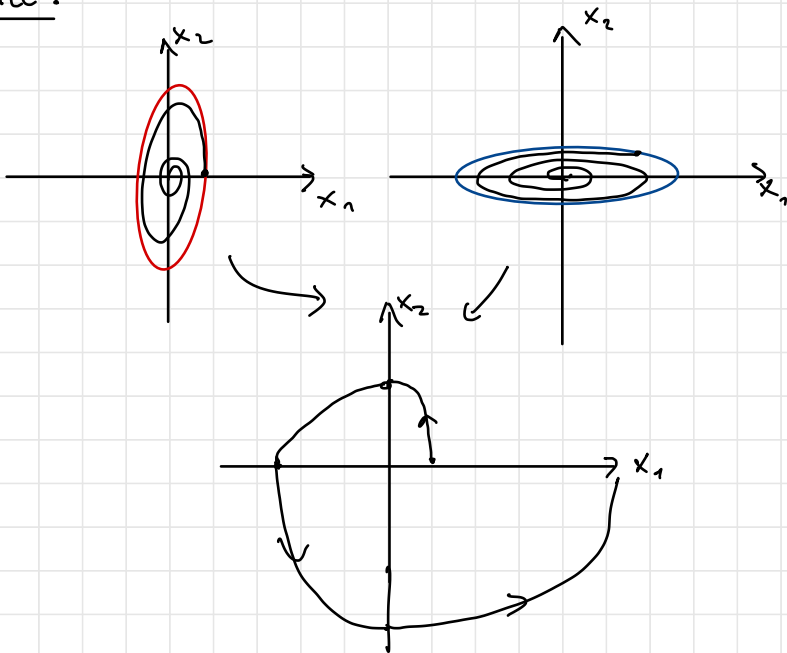


2) The case of time-varying h_k and τ_k

$$x_{k+1} = \Phi(h_k, \tau_k) x_k$$

REMARK: IN CASE OF TIME-VARYING SYSTEMS, THE SCHUR STABILITY OF EVERY MATRIX $\Phi(h, \tau)$ $\forall h, \tau$ IS NEITHER NECESSARY NOR SUFFICIENT FOR THE ASYMPTOTIC STABILITY OF THE SYSTEM.

EXAMPLE:



NUMERICAL EXAMPLES: ASSUME THAT

$$\begin{aligned} x_{k+1} &= F_1 x_k \\ x_{k+2} &= F_2 x_{k+1} \end{aligned}$$

Case 1

F1 =

$$\begin{bmatrix} 0.7384 & 0.5233 \\ 0.9764 & 0.4299 \end{bmatrix} \text{ UNSTABLE}$$

F2 =

$$\begin{bmatrix} 0.2072 & 0.1109 \\ 0.3234 & 0.3752 \end{bmatrix} \text{ AS. STABLE}$$

$$s_1=1.3154, s_2=0.49835, s_m=0.60515 \rightarrow \text{SPECTRAL RADIUS}$$

Case 2

F1 =

$$\begin{bmatrix} 0.7087 & 0.1625 \\ 0.9929 & 0.1136 \end{bmatrix} \text{ AS. STABLE}$$

F2 =

$$\begin{bmatrix} 0.9129 & 0.8518 \\ 0.4817 & 0.8099 \end{bmatrix} \text{ UNSTABLE}$$

$$s_1=0.91101, s_2=1.504, s_m=1.6788$$

Case 3

F1 =

$$\begin{bmatrix} 0.1440 & 0.3379 \\ 0.8506 & 0.2752 \end{bmatrix} \text{ AS. STABLE}$$

F2 =

$$\begin{bmatrix} 0.0060 & 0.4974 \\ 0.8019 & 0.5378 \end{bmatrix} \text{ AS STABLE}$$

$$s_1=0.74967, s_2=0.95719, s_m=0.70359$$

Case 4

F1 =

$$\begin{bmatrix} 0.3299 & 0.8171 \end{bmatrix} \text{ AS. STABLE}$$

THE SYSTEM IS SUCH THAT

The two-step ahead dynamics of the system is described by

$$x_{k+2} = F_1 F_2 x_k \text{ or}$$

$$x_{k+2} = F_2 F_1 x_k$$

THEY HAVE THE SAME EIGENVALUES

IN FACT, IF v IS AN EIGENVECTOR OF $F_1 F_2$ WITH EIGENVALUE λ ,

$$F_2 F_1 v = F_2 (F_1 F_2) v = (F_2 F_1) F_2 v$$

$F_2 v$ is an eigenvector of $F_2 F_1$ with eigenvalue λ !

$\Rightarrow F_1 F_2$ UNSTABLE

0.3421 0.5317

F2 =

$\begin{bmatrix} 0.5211 & 0.1203 \\ 0.7743 & 0.6255 \end{bmatrix}$ AS. STABLE

s1=0.96905, s2=0.88287, sm=1.1985

Case 5

F1 =

$\begin{bmatrix} 0.5603 & 0.3008 \\ 0.6127 & 0.7981 \end{bmatrix}$ UNSTABLE

$\Rightarrow F_1 F_2$ AS. STABLE

F2 =

$\begin{bmatrix} 0.7956 & 0.3511 \\ 0.7811 & 0.0543 \end{bmatrix}$ UNSTABLE

s1=1.1247, s2=1.0666, sm=0.99999

Case 6

F1 =

$\begin{bmatrix} 0.7408 & 0.9469 \\ 0.7376 & 0.5101 \end{bmatrix}$ UNSTABLE

$\Rightarrow F_1 F_2$ UNSTABLE

F2 =

$\begin{bmatrix} 0.7919 & 0.8492 \\ 0.4522 & 0.3904 \end{bmatrix}$ UNSTABLE

s1=1.4691, s2=1.2425, sm=1.8273

A sufficient condition for the stability of the NCS is provided by the following theorem.

Theorem 2: Assume that $\exists z_{\min}, z_{\max}, h_{\min}, h_{\max}$ such that, for all $k \geq 0$:

$$z_{\min} \leq z_k \leq z_{\max}$$

$$h_{\min} \leq h_k \leq h_{\max}$$

The networked control system (NCS) is exponentially stable
IF $\exists P = P^T > 0$ such that, for all $h \in [h_{\min}, h_{\max}]$ and $z \in [z_{\min}, z_{\max}]$,

$$\Phi(h, z)^T P \Phi(h, z) - P < 0$$

REMARKS

- The main idea of Theorem 2 is that it is sufficient that $\Phi(h_k, z_k)$ have a common Lyapunov function to provide stability.

THE COMMON LYAPUNOV FUNCTION IS $V(x_k) = x_k^T P x_k$
IN FACT, FOR $z_k \neq 0$

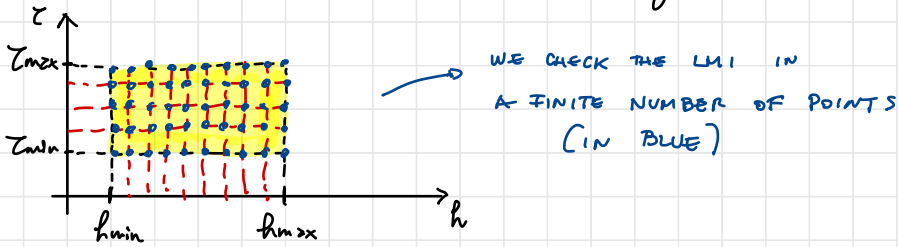
$$\begin{aligned} V(z_{k+1}) &= z_{k+1}^T P z_{k+1} = (\Phi(h_k, z_k) z_k)^T P (\Phi(h_k, z_k) z_k) \\ &= z_k^T \underbrace{\Phi(h_k, z_k)^T P \Phi(h_k, z_k)}_{< P} z_k < z_k^T P z_k = V(z_k) \end{aligned}$$

$$\Rightarrow V(z_{k+1}) < V(z_k) !$$

- In practice we cannot solve a problem with all (h, z) LMIs for all h, z such that

$$h \in [h_{\min}, h_{\max}], z \in [z_{\min}, z_{\max}]$$

To solve this problem in practice, we apply the "gridding" method, i.e., subdividing the intervals $[h_{\min}, h_{\max}]$ and $[z_{\min}, z_{\max}]$ into segments.



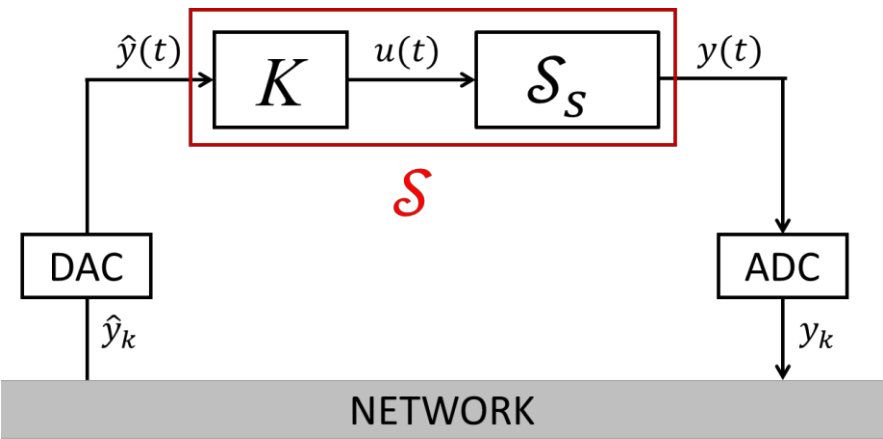
D. MAXIMAL ALLOWABLE TRANSFER INTERVAL (MATI)

MATI IS THE LARGEST VALUE OF h ($\bar{h}$) SUCH THAT,
IF $h_k \leq \bar{h}$, THEN THE NCS IS AS. STABLE,
WITH $z_k \in [z_{\min}, z_{\max}]$.

IT IS IMPORTANT $\rightarrow$ To avoid congestions h should be as
large as possible
 $\rightarrow$ To avoid instabilities.

F. COMPENSATION OF DELAYS

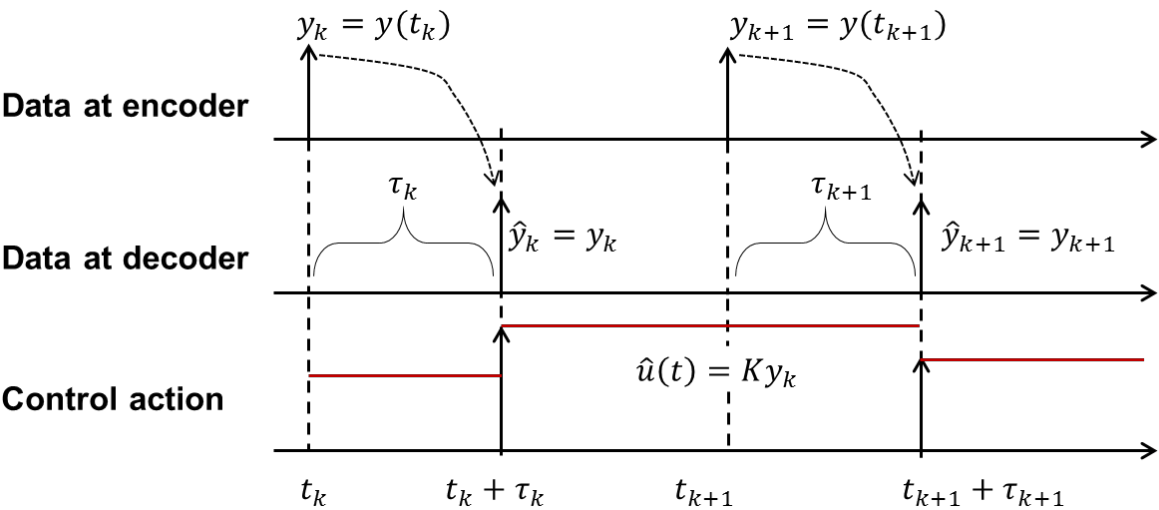
Compensation of delays (controller co-located with actuator)



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

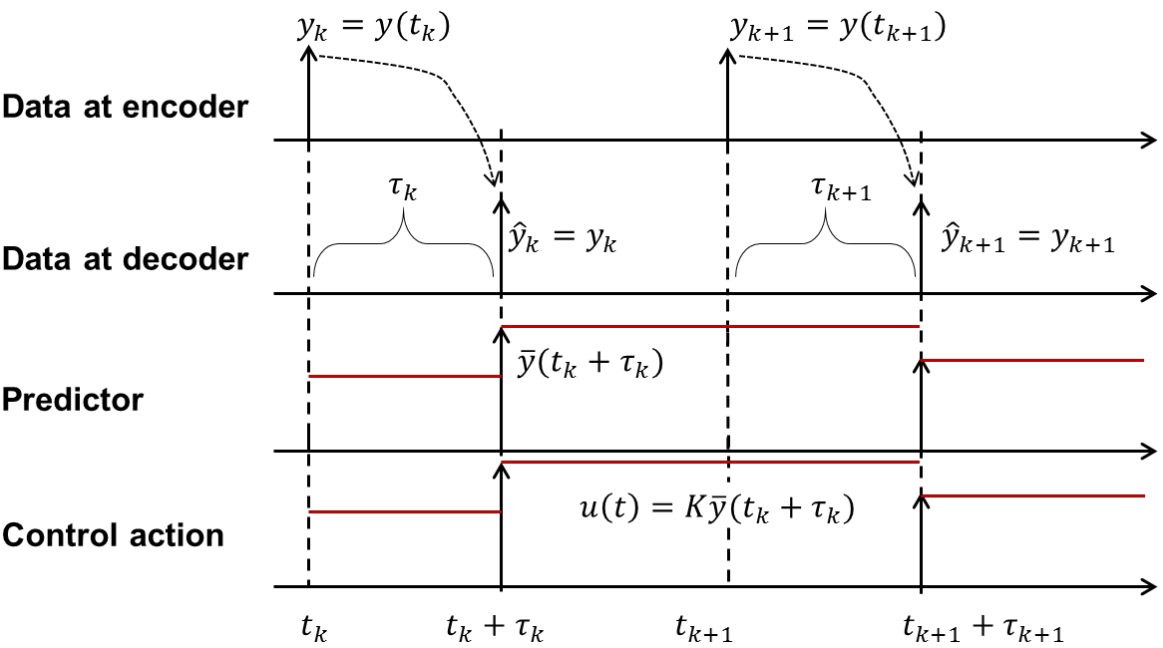
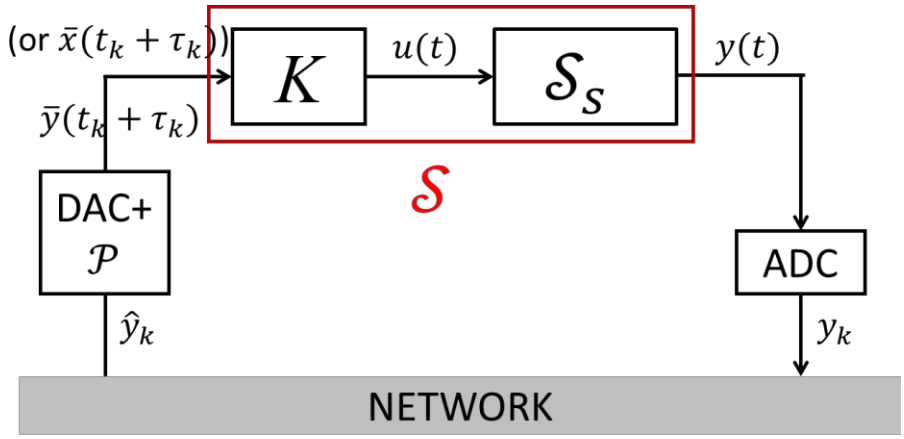
In case time stamps are not used (**unknown delay**)



- $u(t) = Ky(t_k)$ is applied in the interval $[t_k + \tau_k, t_{k+1} + \tau_{k+1})$
- Constant for $h_k^* = h_k + \tau_{k+1} - \tau_k$ s

Compensation of delays (controller co-located with actuator)

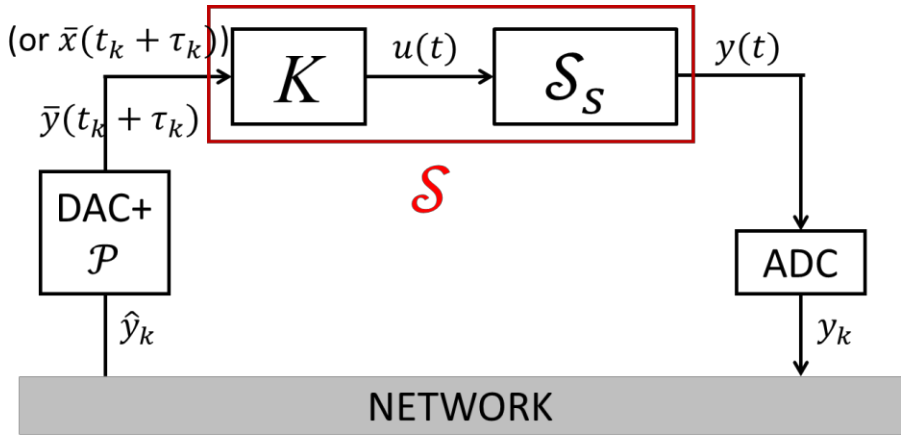
In case **time stamps** are used (**known delay**) a predictor $\mathcal{P}$ can be introduced:



- $u(t) = K\bar{y}(t_k + \tau_k)$ is applied in the interval $[t_k + \tau_k, t_{k+1} + \tau_{k+1})$
- Constant for $h_k^* = h_k + \tau_{k+1} - \tau_k$ s

Compensation of delays (controller co-located with actuator)

In case time stamps are used (**known delay**) a predictor $\mathcal{P}$ can be introduced:



Standard filter. Digital observer of the system state $x_k = x(t_k)$:

$$\bar{x}_{k|k} = e^{A_S h_{k-1}} \bar{x}_{k-1} + e^{A_S (h_{k-1} - \tau_{k-1})} \Gamma(\tau_{k-1}) B_S \hat{u}_{k-2} + \Gamma(h_{k-1} - \tau_{k-1}) B_S \hat{u}_{k-1} \quad \text{prediction}$$

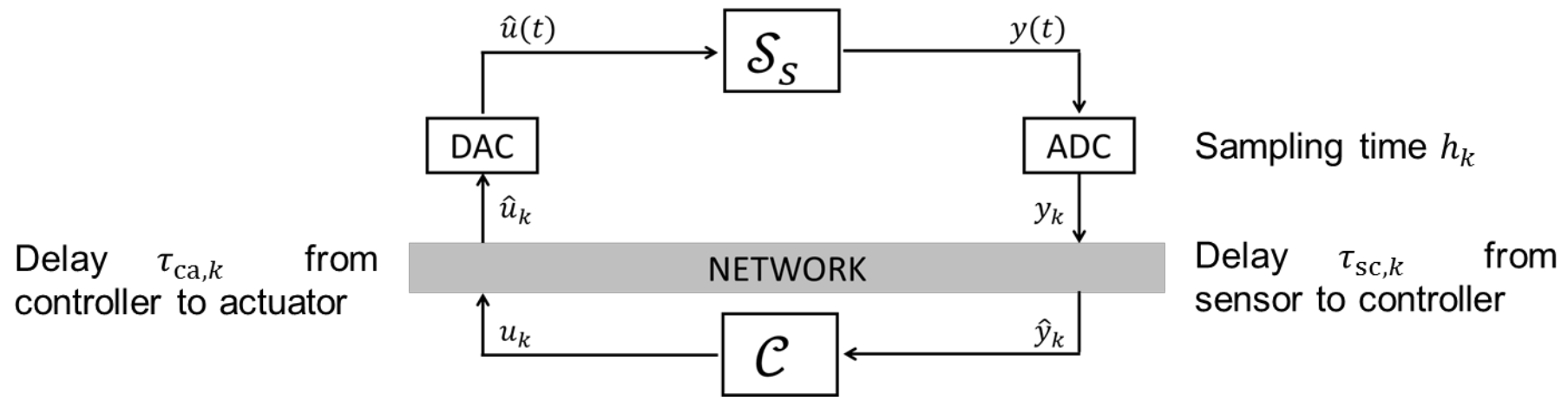
$$\bar{x}_k = \bar{x}_{k|k} + L(y_k - C_S \bar{x}_{k|k}) \quad \text{correction}$$

+ **prediction** (with step τ_k):

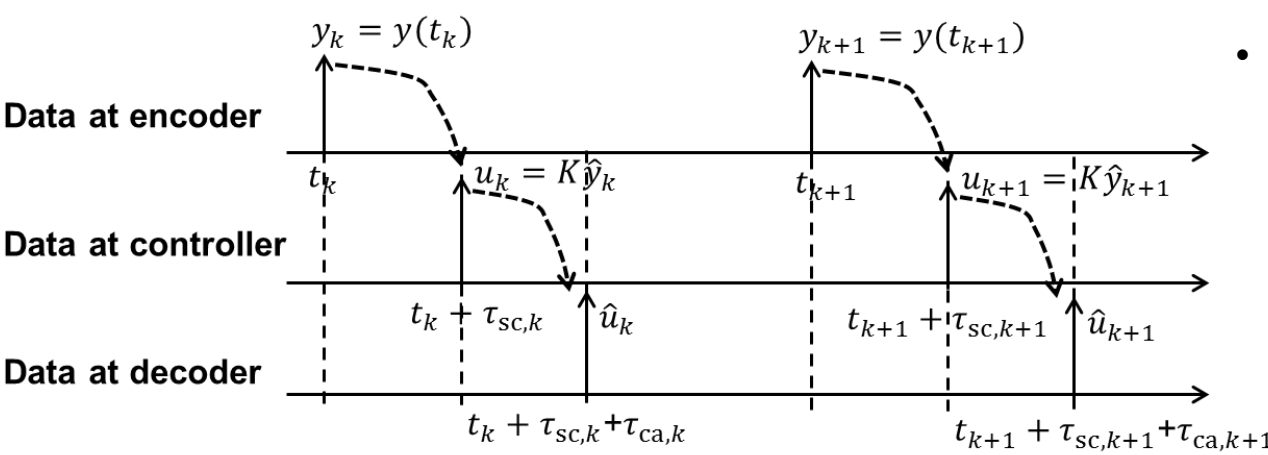
$$\begin{aligned} \bar{x}(t_k + \tau_k) &= e^{A_S \tau_k} \bar{x}_k + \Gamma(\tau_k) B_S \hat{u}_{k-1} \\ \bar{y}(t_k + \tau_k) &= C_S \bar{x}(t_k + \tau_k) \end{aligned}$$

Remark: also state-feedback control laws can be implemented using $\bar{x}(t_k + \tau_k)$.

Compensation of delays (remote controller)



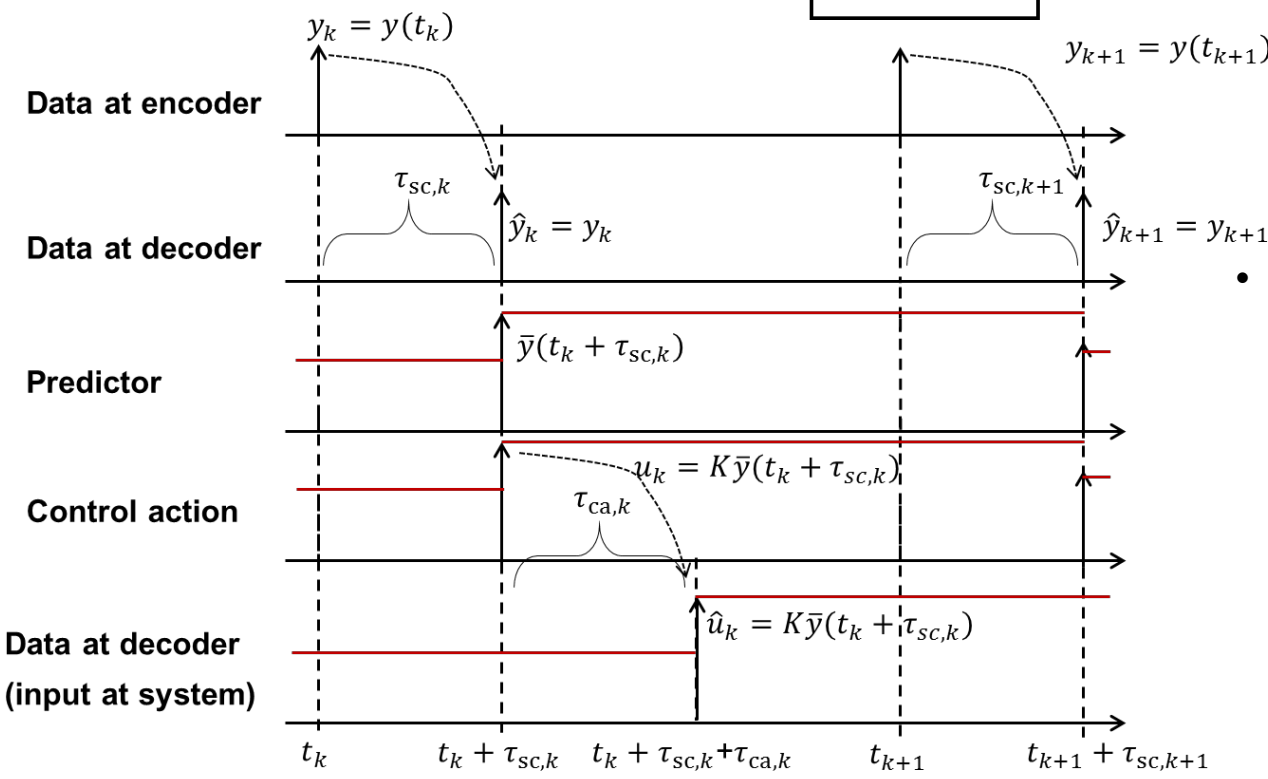
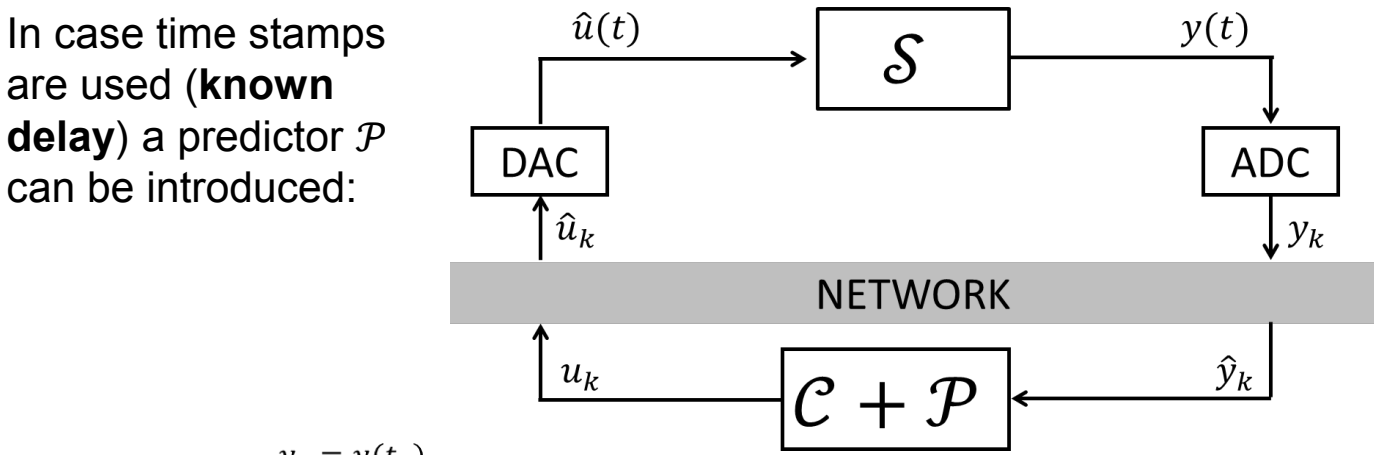
In case time stamps are not used (**unknown delay**)



- $u(t) = Ky(t_k)$ is applied in the interval $[t_k + \tau_{sc,k} +$

Compensation of delays (remote controller)

In case time stamps are used (**known delay**) a predictor $\mathcal{P}$ can be introduced:



- $u(t) = K \bar{y}(t_k + \tau_{sc,k})$ is applied in the interval $[t_k + \tau_{sc,k} + \tau_{ca,k}, t_{k+1} + \tau_{sc,k+1}]$

3. PACKET DROPOUTS

A. CONFIGURATION

$$\hat{y}_k = \begin{cases} y(k) = y_k & \text{IF NO DROPOUT } (\Theta_k = 1) \\ \hat{y}_k = \hat{y}_{k-1} & \text{IF DROPOUT } (\Theta_k = 0) \end{cases}$$

We introduce $\Theta_k = \begin{cases} 1 & \text{IF NO DROPOUT AT TIME } k \\ 0 & \text{IF DROPOUT} \end{cases}$

$$\hat{y}_k = \Theta_k \overset{C x_k}{y_k} + (1 - \Theta_k) \hat{y}_{k-1}$$

B. DYNAMICS OF THE NCS

$$x_{k+1} = e^{A h_k} x_k + e^{A(h_k - \tau_k)} \Gamma(\tau_k) B \hat{y}_{k-1} + \Gamma(h_k - \tau_k) B \hat{y}_k$$

$$= e^{A h_k} x_k + e^{A(h_k - \tau_k)} \Gamma(\tau_k) B \hat{y}_{k-1} +$$

$$\Theta_k \Gamma(h_k - \tau_k) B C x_k + (1 - \Theta_k) \Gamma(h_k - \tau_k) B \hat{y}_{k-1}$$

$$z_k = \begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix}$$

$$z_{k+1} = \begin{bmatrix} x_{k+1} \\ \hat{y}_k \end{bmatrix} = \underbrace{\begin{bmatrix} e^{A h_k} + \Theta_k \Gamma(h_k - \tau_k) B C & (e^{A(h_k - \tau_k)} \Gamma(\tau_k) + (1 - \Theta_k) \Gamma(h_k - \tau_k) B) \\ \Theta_k C & (1 - \Theta_k) I \end{bmatrix}}_{\tilde{\Phi}_k(h_k, \tau_k)} \underbrace{\begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix}}_{z_k}$$

$$\tilde{\Phi}_k(h_k, \tau_k)$$

FROM NOW ON WE CONSIDER ASSUMPTION 1 $\Rightarrow h_k = h, z_k = z \forall k$
 CONSTANT

$z_{k+1} = \tilde{\Phi}_0 z_k \Rightarrow$ SWITCHED SYSTEM $\subseteq$ HYBRID SYSTEM

$$\tilde{\Phi}_1 = \begin{bmatrix} e^{Ah} + \Gamma(h-z)BC & e^{A(h-z)}\Gamma(z)B \\ C & 0 \end{bmatrix} = \tilde{\Phi}(h, z)$$

Assumption: $\tilde{\Phi}(h, z)$
Schur-stable

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{Ah} \\ 0 \end{bmatrix} \rightarrow F$$

$$\begin{bmatrix} (e^{A(h-z)}\Gamma(z) + \Gamma(h-z))B \\ I \end{bmatrix}$$

$$\begin{aligned} & e^{A(h-z)} \int_0^z e^{Av} dv + \int_0^{h-z} e^{Av} dv \\ & \int_0^z e^{A(z-v)} dv \quad \int_z^h e^{A(h-v)} dv \\ & = \int_0^z e^{A(h-z)} e^{A(z-v)} dv + \int_z^h e^{A(h-v)} dv \\ & = \int_0^z e^{A(h-v)} dv + \int_z^h e^{A(h-v)} dv \\ & = \int_0^h e^{A(h-v)} dv = \int_0^h e^{Av} dv \end{aligned}$$

$$\int_0^h e^{Av} dv B = \Gamma(h)B = G$$

$$\Rightarrow \tilde{\Phi}_0 = \begin{bmatrix} F & G \\ 0 & I \end{bmatrix} \quad \text{TRIANGULAR MATRIX}$$

↳ Eigenvalues of $\tilde{\Phi}_0 = \text{eigs}(F) \cup \{1, \dots, 1\}$

TWO CASES:

- IF $\mathcal{S}$ IS UNSTABLE $\Rightarrow \tilde{\Phi}_0$ UNSTABLE
- IF $\mathcal{S}$ IS AS. STABLE $\Rightarrow \tilde{\Phi}_0$ IS NOT ASYMPTOTICALLY STABLE (SPECTRAL RADIUS ≥ 1)

LIMIT CASES: • ALWAYS COMMUNICATION ($\theta_k = 1 \quad \forall k$) ($r = 0$)

$$z_{k+1} = \tilde{\Phi}_1 z_k \quad (\text{AS. STABLE})$$

• NEVER COMMUNICATION ($\theta_k = 0 \quad \forall k$) ($r = 1$)

$$z_{k+1} = \tilde{\Phi}_0 z_k \quad (\text{NOT AS. STABLE})$$

For characterizing

the dropouts there are two frameworks

→ DETERMINISTIC - DROPOUTS OCCUR REGULARLY
 → STOCHASTIC - θ_k IS A BERNOULLI PROCESS

C. DETERMINISTIC FRAMEWORK: ASYMPTOTIC DROPOUT RATE

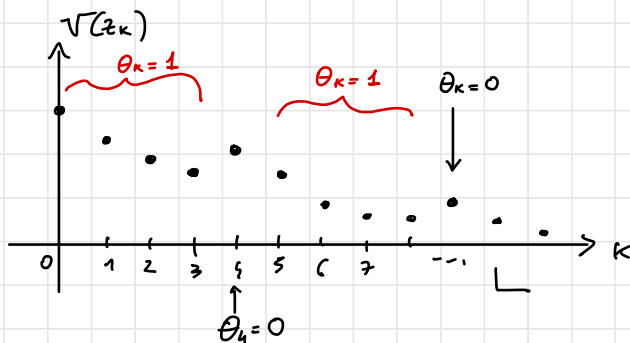
Asymptotic dropout rate:

$$r = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{k=k_0}^{k_0+T-1} (1 - \theta_k) \quad \text{constant for all } k_0 \geq 0$$

1) STABILITY OF THE NCS

To analyze the stability of the NCS under packet dropouts consider $V(x_k)$, the Lyapunov function of the system when $\theta_k = 1$ (with no dropouts): $V(x_k) = x_k^T P x_k$. By definition $V(z_{k+1}) < V(z_k)$ if $\theta_k = 1$. On the other hand $V(z_{k+1}) \geq V(z_k)$ if $\theta_k = 0$.

Graphically, we can represent this situation as in this figure



• When $\theta_k = 1$: $V(x_{k+1}) = x_k^T \tilde{F}_1^T P \tilde{F}_1 x_k \leq \beta_1 x_k^T P x_k \leq \beta_1 V(x_k)$

$\tilde{F}_1^T P \tilde{F}_1 < P \Rightarrow \tilde{F}_1^T P \tilde{F}_1 \leq \beta_1 P \quad \beta_1 \in [0, 1]$

• When $\theta_k = 0$: $\tilde{F}_0^T P \tilde{F}_0 \leq \beta_0 P, \beta_0 > 1 \Rightarrow V(x_{k+1}) \leq \beta_0 V(x_k)$

• In general:

$$V(z_k) \leq \beta_{\theta_{k-1}} V(z_{k-1}) \leq \beta_{\theta_{k-1}} \beta_{\theta_{k-2}} \dots \beta_{\theta_0} V(z_0)$$

For stability, it is sufficient that this term is < 1

$\beta_0^{kr} \beta_1^{k(1-r)} = \left(\beta_0^r \beta_1^{(1-r)} \right)^k < 1 \text{ for stability!}$

THIS COMES FROM THE FACT THAT, IN k STEPS $\rightarrow$ kr TIMES DROPOUTS $\Rightarrow \theta_k = 0, \beta_{\theta_k} = \beta_0$
 $\rightarrow k(1-r)$ TIMES COMMUNICATION $\Rightarrow \theta_k = 1 \Rightarrow \beta_{\theta_k} = \beta_1$

Theorem: Assume that $\exists P = P^T > 0$ and scalars β_0, β_1 s.t.

- $\beta_0^r \beta_1^{1-r} < 1$
- $\tilde{Z}_0^T P \tilde{Z}_0 \leq \beta_0 P$
- $\tilde{Z}_1^T P \tilde{Z}_1 \leq \beta_1 P$

Then, the NCS is exponentially stable.

2) COMPUTATION OF THE MAXIMAL DROPOUT RATE

Now we address the following question: given a NCS, what is the maximal dropout rate $\bar{r}$ such that the NCS is as. stable $\forall r < \bar{r}$?

Assume that $\exists P = P^T > 0$ and scalars β_0, β_1 s.t.

$$\tilde{Z}_0^T P \tilde{Z}_0 \leq \beta_0 P \quad (\beta_0 \geq 1)$$

$$\tilde{Z}_1^T P \tilde{Z}_1 \leq \beta_1 P \quad (\beta_1 \in [0, 1])$$

To find $\bar{r}$ we use the inequality $\beta_0^r \beta_1^{1-r} < 1$, given β_0, β_1 .

$$\log(\beta_0^r \beta_1^{1-r}) < \log 1$$

$\Downarrow$

$$r \log \beta_0 + (1-r) \log \beta_1 < 0$$

$\Downarrow$

$$r \underbrace{(\log(\beta_0))}_{> 1} - \underbrace{(\log(\beta_1))}_{< 1} < -\log(\beta_1)$$

$\begin{matrix} > 0 & & & & \\ & \searrow & & \nearrow & \\ & & & & \end{matrix}$

This results in: $r < \frac{-\log(p_1)}{\log(p_0) - \log(p_1)} = \frac{1}{1 - \frac{\log(p_0)}{\log(p_1)}} = \bar{r}$

↓
> 0

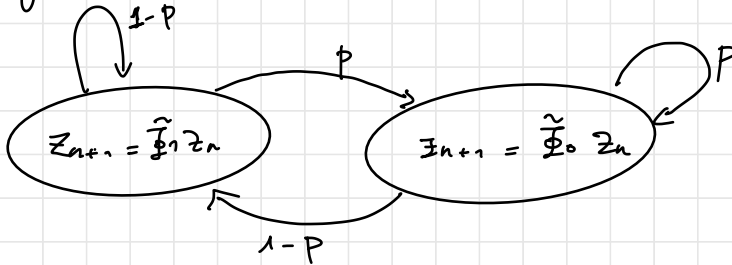
Note that $\bar{r} \in (0, 1]$

D. STOCHASTIC DROPOUTS

In this case, the assumption is that θ_k is a Bernoulli stochastic process, i.e.,

- $\theta_k = 0$ with probability p
- $\theta_k = 1$ with probability $1-p$

The NCS dynamics can be graphically represented by the graph:



$\Rightarrow$ Markov Jump Linear System (MJLS)

Theorem: The NCS is mean-square stable if $\exists Z = Z^T > 0$ such that

$$\begin{bmatrix} Z & \sqrt{p} (\tilde{\Phi}_0 Z)^T & \sqrt{1-p} (\tilde{\Phi}_1 Z)^T \\ \sqrt{p} (\tilde{\Phi}_0 Z) & Z & 0 \\ \sqrt{1-p} (\tilde{\Phi}_1 Z) & 0 & Z \end{bmatrix} > 0$$

Mean square stability:

$$\bullet \bar{z}_k = \mathbb{E}[z_k] \xrightarrow[k \rightarrow \infty]{} 0$$

$$\bullet \mathbb{E}[z_k z_k^T] = \bar{z}_k \bar{z}_k^T + \text{var}(z_k) \xrightarrow[k \rightarrow \infty]{} 0$$

Example 1

The dynamics of the scalar system $\mathcal{S}_S$ is

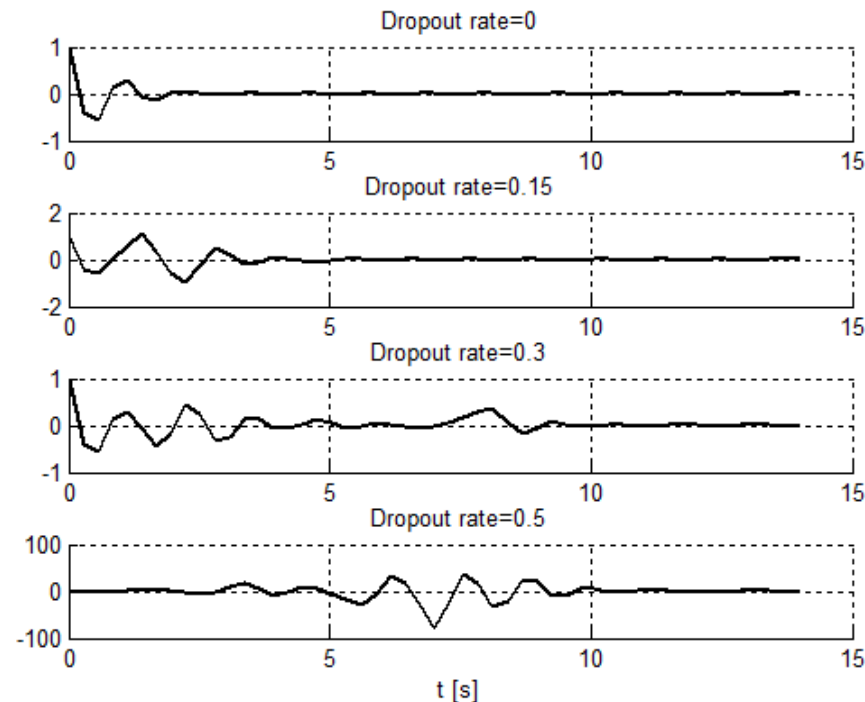
$$\begin{cases} \dot{x}_S(t) &= u(t) \\ y(t) &= x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \Phi(h, \tau) = \begin{bmatrix} 1 + K(h - \tau) & K\tau \\ 1 & 0 \end{bmatrix}$$
$$\tilde{\Phi}_0 = \begin{bmatrix} 1 & Kh \\ 0 & I \end{bmatrix}$$

$$\bar{r} = 0.2802, \bar{p} = 0.23$$

($K = -5$)



Example 2

The dynamics of the scalar system $\mathcal{S}_s$ is

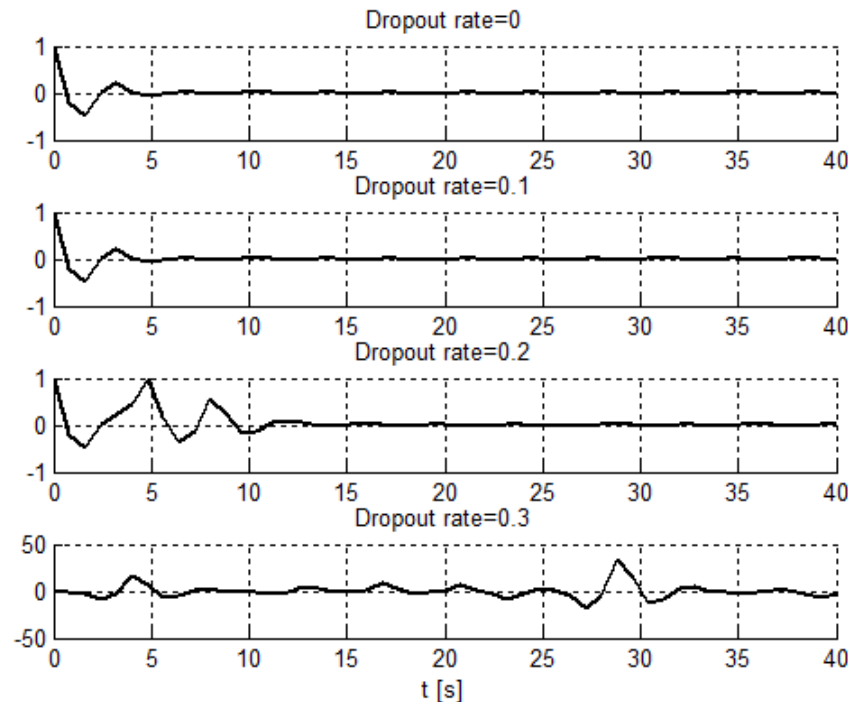
$$\begin{cases} \dot{x}_s(t) &= x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} e^h + K(e^{(h-\tau)} - 1) & Ke^{(h-\tau)}(e^\tau - 1) \\ 1 & 0 \end{bmatrix}, \tilde{\Phi}_0 = \begin{bmatrix} e^h & K(e^h - 1) \\ 0 & 1 \end{bmatrix}$$

$$\bar{r} = 0.2224, \bar{p} = 0.08$$

$(k = -2)$



Example 3

The dynamics of the scalar system $\mathcal{S}_S$ is

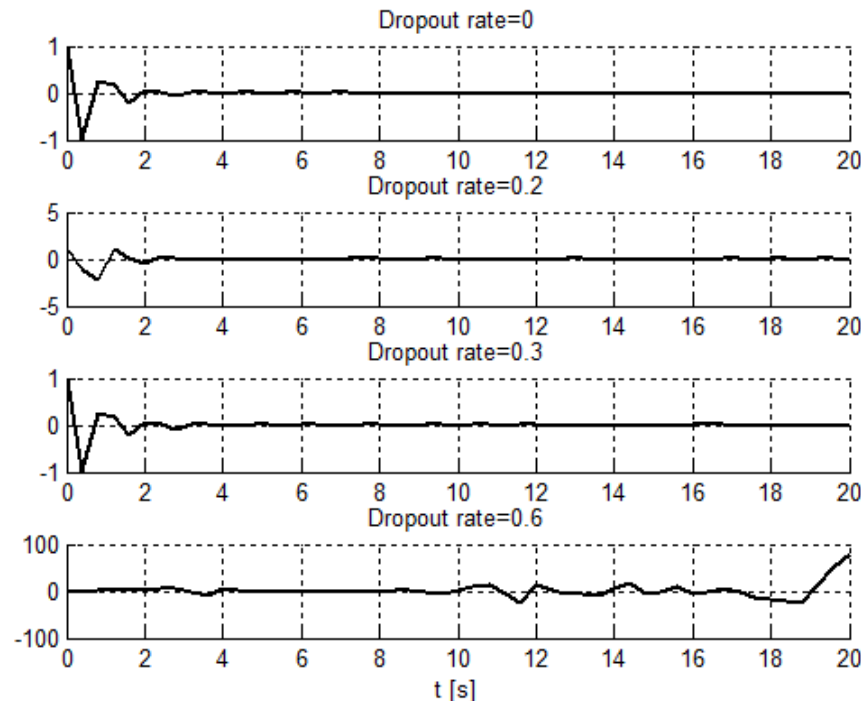
$$\begin{cases} \dot{x}_S(t) &= -x_S(t) + u(t) \\ y(t) &= x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} e^{-h} + K(1 - e^{-(h-\tau)}) & Ke^{-(h-\tau)}(1 - e^{-\tau}) \\ 1 & 0 \end{bmatrix}, \tilde{\Phi}_0 = \begin{bmatrix} e^{-h} & K(1 - e^{-h}) \\ 0 & 1 \end{bmatrix}$$

$$\bar{r} = 0.2975, p = 0.19$$

(k = -5)



Example 4

The dynamics of the 2x2 system $\mathcal{S}$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}, A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = [-0.1 \ 1]$$

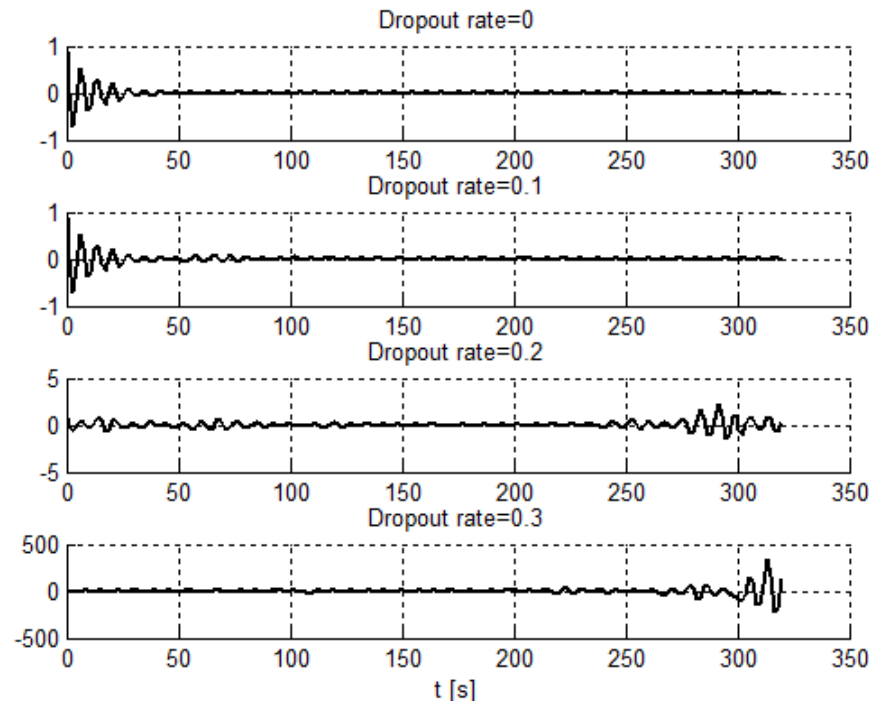
and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} e^{A_S h} + A_S^{-1}(e^{A_S(h-\tau)} - I)B_S K C_S & e^{A_S(h-\tau)} A_S^{-1}(e^{A_S \tau} - I)B_S K \\ C_S & 0 \end{bmatrix}$$

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{A_S h} & A_S^{-1}(e^{A_S h} - I)B_S K \\ 0 & 1 \end{bmatrix}$$

$$\bar{r} = 0.1119, \bar{p} = 0.11$$

(k = -5)



EXERCISES

1) $\dot{x} = -x + u$

A) $\dot{x} = -x + Kx \quad u = Kx$

$$\downarrow$$

$$(-1+K)x$$

$$\downarrow$$

$$\lambda$$

$$-1+K < 0$$

$$K < 1$$

B) $h = \log 2.5$

$$u_n = Kx_n$$

$$x_{n+1} = f x_n + g u_n$$

$$\downarrow$$

$$= 0.4x_n + 0.6u_n$$

$$\downarrow$$

$$= (0.4 + 0.6K)x_n$$

$$\downarrow$$

$$\lambda$$

$$f = e^{-h} = e^{-\log 2.5}$$

$$\downarrow$$

$$= e^{\log(2.5)^{-1}} = 0.4$$

$$g = \frac{-b}{a}(e^{ah} - 1)$$

$$\downarrow$$

$$= 1 - 0.4 = 0.6$$

$$|\lambda| < 1$$

$$\begin{cases} 0.4 + 0.6K < 1 \Rightarrow K < 1 \\ 0.4 + 0.6K > -1 \Rightarrow 0.6K > -1.4 \end{cases}$$

$$\frac{3}{5}K > -\frac{7}{5}$$

$\Rightarrow$

$$-\frac{7}{3} < K < 1$$

$$\downarrow$$

$$K > -\frac{7}{3}$$

c) 2) $\alpha = \log 1.5$

$$\Phi(k, z) = \begin{bmatrix} e^{a(h-z)} + \Gamma(h-z) b k c & e^{a(h-z)} \Gamma(z) b k \\ c & 0 \end{bmatrix}$$

The Schur stability of $\Phi(k, z)$ guarantee as. stability of the NCS

$$\Gamma(\mu) = -\frac{1}{a}(1 - e^{a\mu}) \quad (a \neq 0)$$

$$= \int_0^\mu e^{av} dv$$

$$e^{ah} = e^{-\log 2.5} = 0.4 = \frac{2}{5}$$

$$e^{a(h-c)} = e^{-(\log 2.5 - \log 1.5)} = e^{-\log \frac{2.5}{1.5}} = e^{-\log \frac{5}{3}}$$

$$= e^{\log \frac{3}{5}} = \frac{3}{5}$$

$$\Gamma(h-c) = -\frac{1}{a}\left(1 - \frac{3}{5}\right) = \frac{2}{5}$$

$$e^{ay} = e^{-\log 1.5} = e^{\log(\frac{2}{3})} = \frac{2}{3}$$

$$\Gamma(\mu) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\Rightarrow \Phi(h, c) = \begin{bmatrix} \frac{2}{5} + \frac{2}{5}K & \frac{3}{5}\frac{1}{3}K \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{2}{5}(1+K) & \frac{1}{5}K \\ 1 & 0 \end{bmatrix}$$

We apply the

Jury criterion: 1) $P_{\frac{1}{2}}(\lambda) = \det \begin{bmatrix} \lambda - \frac{2}{5}(1+K) & -\frac{1}{5}K \\ -1 & \lambda \end{bmatrix}$

$$= \lambda^2 - \underbrace{\frac{2}{5}(1+K)}_a \lambda - \underbrace{\frac{1}{5}K}_b$$

2) The Schur stability of $\Phi(h, c)$ is guaranteed if and only if

$$\begin{aligned} -1 < -b < 1 \\ -1 < \frac{1}{5}K < 1 \end{aligned} \quad \Bigg\| \Rightarrow \quad \begin{aligned} -1 < \frac{1}{5}K < 1 \\ -1 + \frac{1}{5}K < \frac{2}{5}(1+K) < 1 - \frac{1}{5}K \end{aligned}$$

$$\begin{cases} -5 < K < 5 \\ \frac{2}{5} + \frac{2}{5}K < 1 - \frac{1}{5}K \Rightarrow \frac{3}{5}K < 1 - \frac{2}{5} = \frac{3}{5} \Rightarrow K < 1 \\ \frac{2}{5} + \frac{2}{5}K > -1 + \frac{1}{5}K \Rightarrow \frac{1}{5}K > -\frac{7}{5} \\ \hookrightarrow K > -7 \end{cases}$$

$$-5 < K < 1$$

$$b) \quad z = \log 2$$

$$e^{ah} = e^{-\log 2.5} = 0.4$$

$$e^{a(h-z)} = e^{-(\log 2.5 - \log 2)} = e^{-\log \frac{5}{2} \cdot \frac{1}{2}} = e^{\log \frac{4}{5}} = \frac{4}{5}$$

$$\Gamma(h-z) = 1 - \frac{4}{5} = \frac{1}{5}$$

$$e^{ay} = e^{-\log 2} = \frac{1}{2}$$

$$\Gamma(z) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\mathbb{L}(h, z) = \begin{bmatrix} \frac{2}{5} + \frac{1}{5}K & \frac{4}{5} \cdot \frac{1}{2} \cdot K \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{2}{5} + \frac{1}{5}K & \frac{2}{5}K \\ 1 & 0 \end{bmatrix}$$

$$p(\lambda) = \left(\lambda - \left(\frac{2}{5} + \frac{1}{5}K \right) \right) \lambda - \frac{2}{5}K = \lambda^2 - \left(\frac{2}{5} + \frac{1}{5}K \right) \lambda - \frac{2}{5}K$$

$$\begin{cases} -1 < \frac{2}{5}K < 1 \\ -1 + \frac{2}{5}K < \frac{2}{5} + \frac{1}{5}K < 1 - \frac{2}{5}K \end{cases} \quad \Rightarrow \text{Jury criterion}$$

$$\begin{cases} -\frac{5}{2} < k < \frac{5}{2} \\ \frac{1}{5}k + \frac{2}{5}k < 1 - \frac{2}{5} \rightarrow \frac{3}{5}k < \frac{3}{5} \rightarrow k < 1 \\ \frac{1}{5}k < \frac{2}{5} + 1 \rightarrow \frac{1}{5}k < \frac{7}{5} \rightarrow k < 7 \end{cases}$$

$$-\frac{5}{2} < k < 1$$

D)

$$\Phi(h, z) = \begin{bmatrix} e^{ah} + \Gamma(h-z)bk & e^{a(h-z)}\Gamma(z)bk \\ c & 0 \end{bmatrix}$$

$k = -1$, $z = \log 1.5$, MATI: MAX VALUE OF h FOR ASYMPTOTIC STABILITY

$e^{ah} = e^{-h} = f$ *We consider it as our "new variable"*

$e^{a(h-z)} = e^{ah} e^{-az} = e^{-h} e^{-z} = f e^{\log 1.5} = \frac{3}{2}f$

$\Gamma(h-z) = 1 - \frac{3}{2}f$

$e^{az} = e^{-z} = e^{-\log 1.5} = \frac{2}{3}$

$\Gamma(z) = 1 - \frac{2}{3} = \frac{1}{3}$

$$\Rightarrow \Phi(f, z) = \begin{bmatrix} f - (1 - \frac{3}{2}f) & -\frac{3}{2}f \cdot \frac{1}{3} \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{5}{2}f - 1 & -\frac{1}{2}f \\ 1 & 0 \end{bmatrix}$$

$$p(\lambda) = \det(\lambda I - \tilde{F}) = \lambda^2 - \underbrace{\left(\frac{5}{2}f - 1\right)}_a \lambda + \underbrace{\frac{1}{2}f}_b$$

JURY
CRITERION

$$\begin{cases} -1 < \frac{1}{2}f < 1 \Rightarrow -2 < f < 2 \\ \frac{5}{2}f - 1 < 1 + \frac{1}{2}f \Rightarrow \frac{4}{2}f < 2 \Rightarrow f < 1 \\ \frac{5}{2}f - 1 > -1 - \frac{1}{2}f \Rightarrow \frac{6}{2}f > 0 \Rightarrow f > 0 \end{cases}$$

$$0 < f < 1$$

$$\Rightarrow \text{No restrictions on } h \Rightarrow \bar{h} = +\infty$$

$$\text{E) } h = \log 2.5, \quad z = 0, \quad k = -1$$

$$\boxed{\theta_k = 1} \Rightarrow \tilde{x}_{k+1} = \begin{bmatrix} x_{k+1} \\ y_k \end{bmatrix} = \tilde{F}(h, z) \tilde{x}_k$$

$$\tilde{F}(h, z) = \begin{bmatrix} e^{ah} + \Gamma(h-z)bk & e^{a(h-z)}\Gamma(z)bk \\ 1 & 0 \end{bmatrix}$$

$$e^{ah} = 0.4 \quad \Gamma(h-z) = \Gamma(h) = 1 - 0.4 = 0.6$$

$$e^{a(h-z)} = e^{ah} = 0.4 \quad \Gamma(z) = 1 - e^{-z} = 0$$

$$\Phi(h, z) = \begin{bmatrix} 0.4 + 0.6K & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -0.2 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\cdot \theta_K = 0$$

$$\Rightarrow Z_{k+1} = \tilde{\Phi}_0 Z_k + \underbrace{\Gamma(h)bk}_{G} \Rightarrow \text{W DISCRETE-TIME} \\ X_{k+1} = FX_k + GUK$$

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{ah} & \Gamma(h-z)bk + e^{a(h-z)}\Gamma(z)bk \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4 & -0.6 \\ 0 & 1 \end{bmatrix}$$

$$Z_2 = \Phi_1 Z_1$$

$$Z_4 = \Phi_1 \Phi_0 Z_2$$

$$Z_3 = \Phi_0 \Phi_1 Z_1$$

$$Z_3 = \Phi_0 Z_2$$

$$Z_6 = \Phi_1 \Phi_0 Z_4$$

$$Z_5 = \Phi_0 \Phi_1 Z_3$$

$$Z_4 = \Phi_1 Z_3$$

$$Z_5 = \Phi_0 Z_4$$

THE TRANSITION MATRIX

$$\text{IS } \boxed{\Phi_1 \Phi_0}$$

THE TRANSITION

$$\text{MATRIX IS } \boxed{\Phi_0 \Phi_1}$$

THEY HAVE THE SAME EIGENVALUES

$$\Phi_1 \Phi_0 = \begin{bmatrix} -0.2 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0.4 & -0.6 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{25} & \frac{3}{25} \\ \frac{2}{5} & -\frac{3}{5} \end{bmatrix}$$

$$\begin{aligned}
 p(\lambda) &= \det(\lambda I - \tilde{F}_1 \tilde{F}_0) = \det \begin{bmatrix} \lambda + \frac{2}{25} & -\frac{3}{25} \\ -\frac{2}{5} & \lambda + \frac{3}{5} \end{bmatrix} \\
 &= \lambda^2 + \left(\frac{2}{25} + \frac{3}{5}\right)\lambda + \frac{6}{125} - \frac{6}{125} \\
 &= \lambda^2 + \frac{17}{25}\lambda = 0 \quad \begin{array}{l} \nearrow \lambda = 0 \\ \searrow \left| \lambda \neq 0 \right| - \frac{17}{25} < 1 \end{array}
 \end{aligned}$$

$\Rightarrow$ AS. STABILITY OF THE NCS UNDER DROPOUTS!